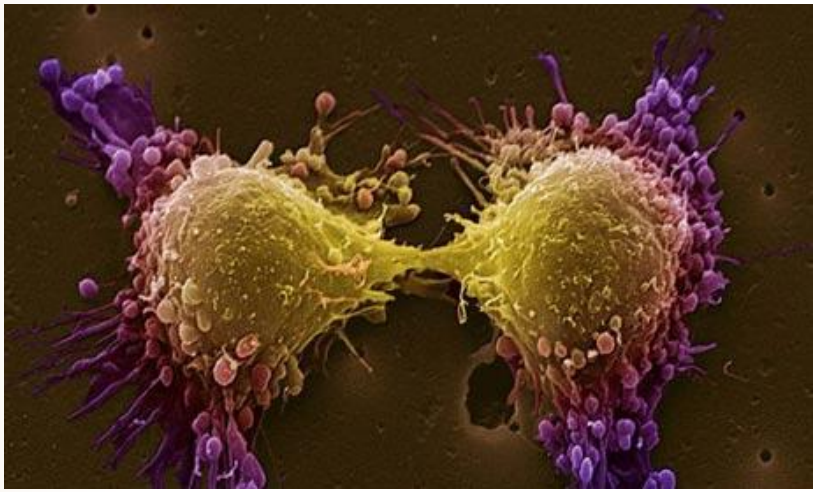
Cell division and function

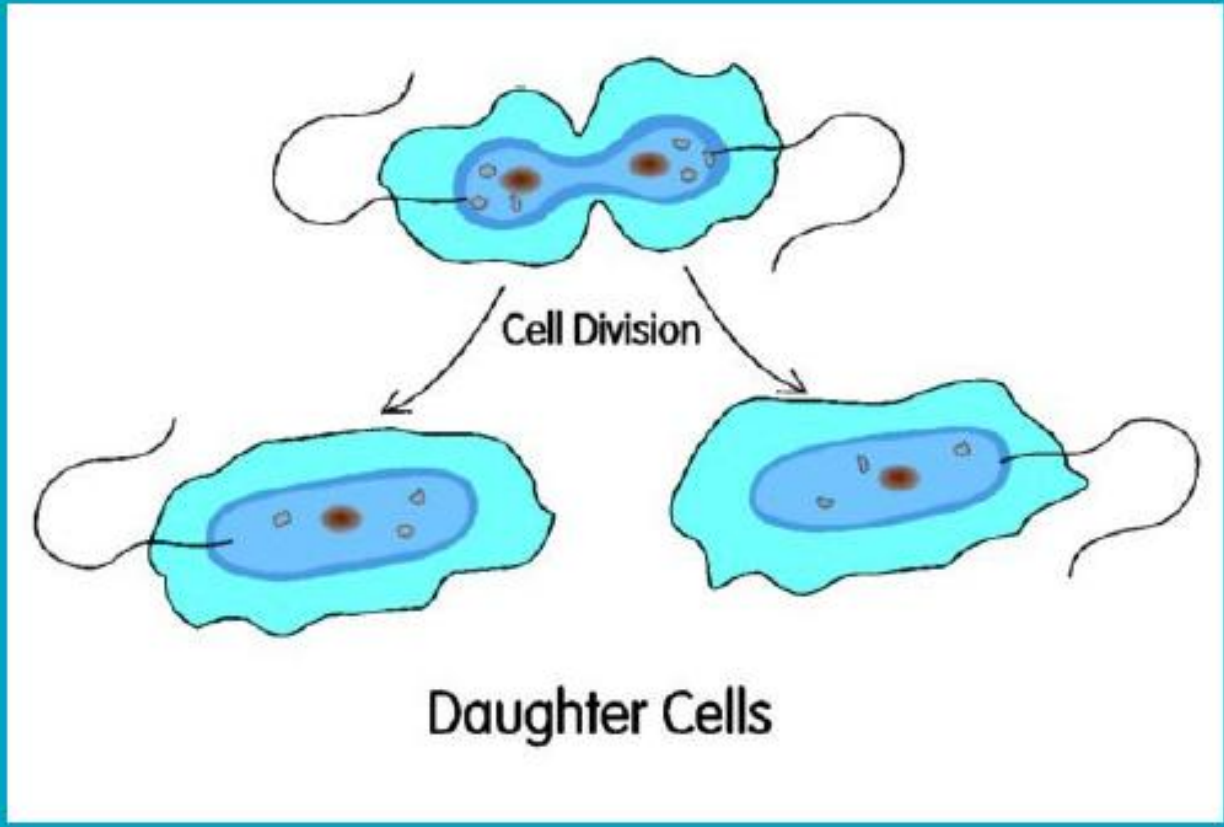


Cell Division

What is it?

Why do Cells do it?

Why is it important to me?



Why do cells undergo divisions or Why cell divides?

1). To prevent DNA overload: If cell grows without limit, "information crisis" would develop. DNA cannot meet the demand of increasing cell size. Therefore, cell divides to increase the number rather than increasing the size.

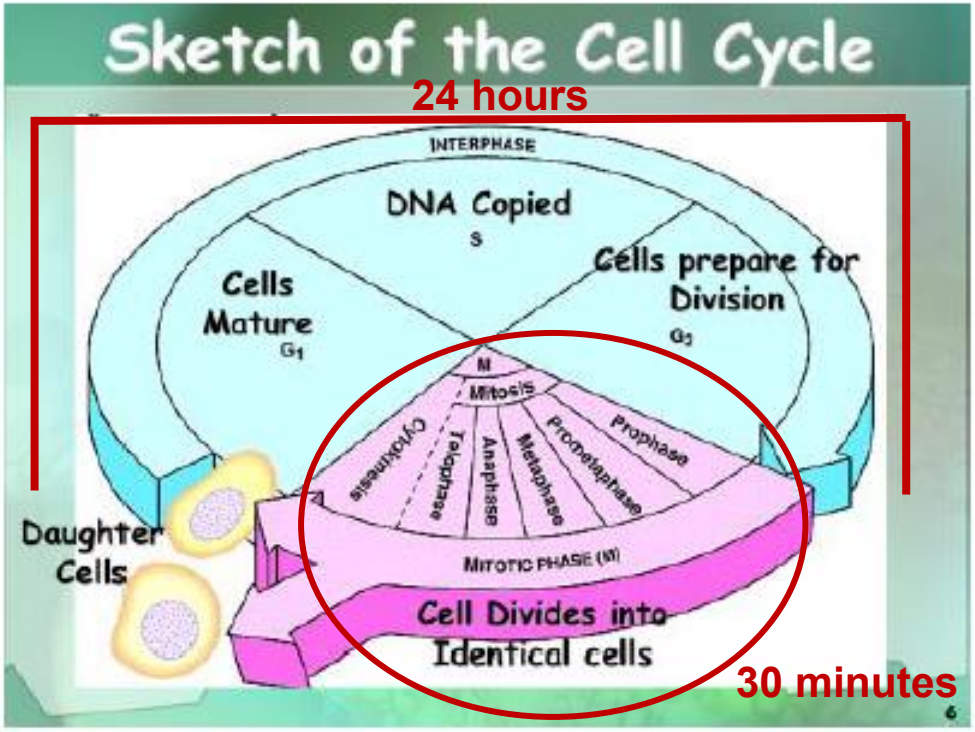
2). For efficient exchange of materials: If cell is too large, exchange of materials through cell membrane such as food and oxygen, and removal of waste products, will occur slowly and cell will die either due to starvation or toxic wastes accumulation.

3) Growth and reproduction: Organisms need to grow which is fulfilled by increasing cell numbers rather than increasing cell size. For example, a single celled zygote (sperm and ova fertilization) develops by **number of cell divisions to form different parts of body and ultimately a full organism.**

4) Tissue repair and regeneration: Dead and injured cells can be replaced by cell divisions.

What is cell cycle

The **cell cycle**, or **cell-division cycle**, is the series of events that take place in a cell **that cause it to divide into daughter cells**. These events include the duplication of its DNA (DNA replication) and some of its organelles, and subsequently the partitioning of its cytoplasm and other components into daughter cells in a process called cell division



Five Phases of the Cell Cycle

- ✓G_1 - primary growth phase
- ✓S - synthesis; DNA replicated
- ✓G_2 - secondary growth phase

collectively these 3 stages are called interphase

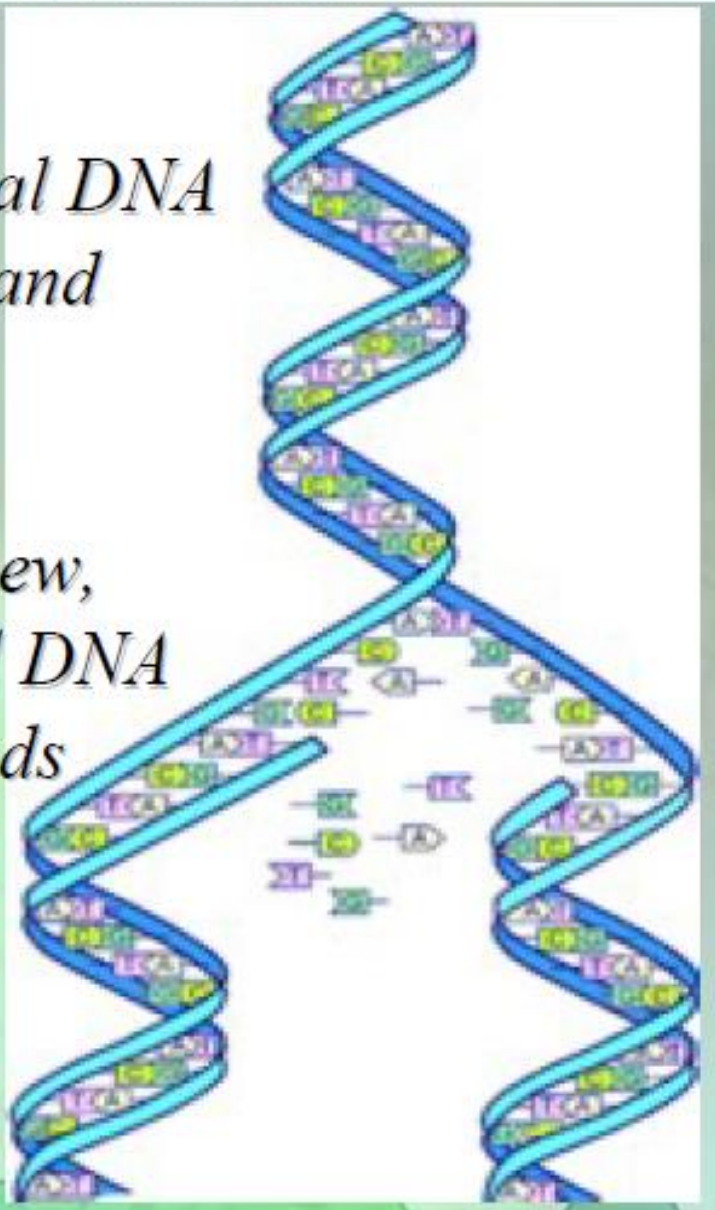
- ✓M - mitosis
- ✓C - cytokinesis

Stages of Interphase

Interphase stage of cell cycle	Important events in a particular phase
G1 (Gap 1)- primary growth phase	Increase in cell size, synthesis of proteins, carbohydrates and lipids, Synthesis of RNA, synthesis of enzymes and energy molecules. **(Time duration: generally 5-6 hours, sometimes 9 hours depending upon the cell type and organism)**
S (Synthesis)	Replication of DNA, synthesis of histones and formation of new nucleosomes. **(Time duration: 10-12 hours)**
G2 (Gap 2) - secondary growth phase	Synthesis of spindle proteins, duplication of mitochondria, , synthesis and storage of ATP molecules for M-phase, Damaged DNA is repaired. **(Time duration: 4-6 hours)**

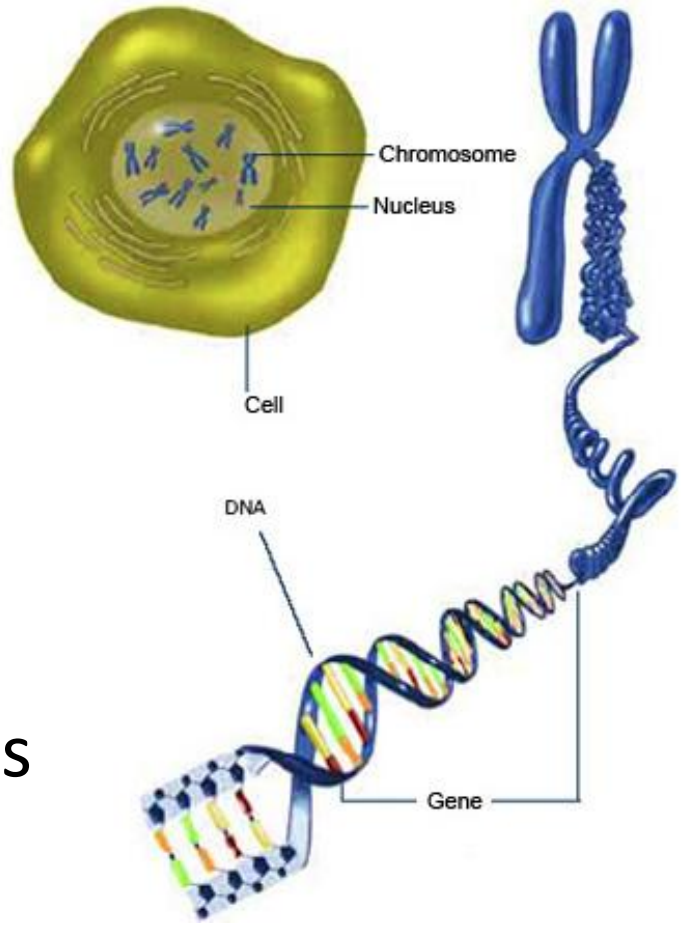
DNA Replication

- DNA must be copied or **replicated** before cell division
- Each new cell will then have an **identical copy** of the DNA

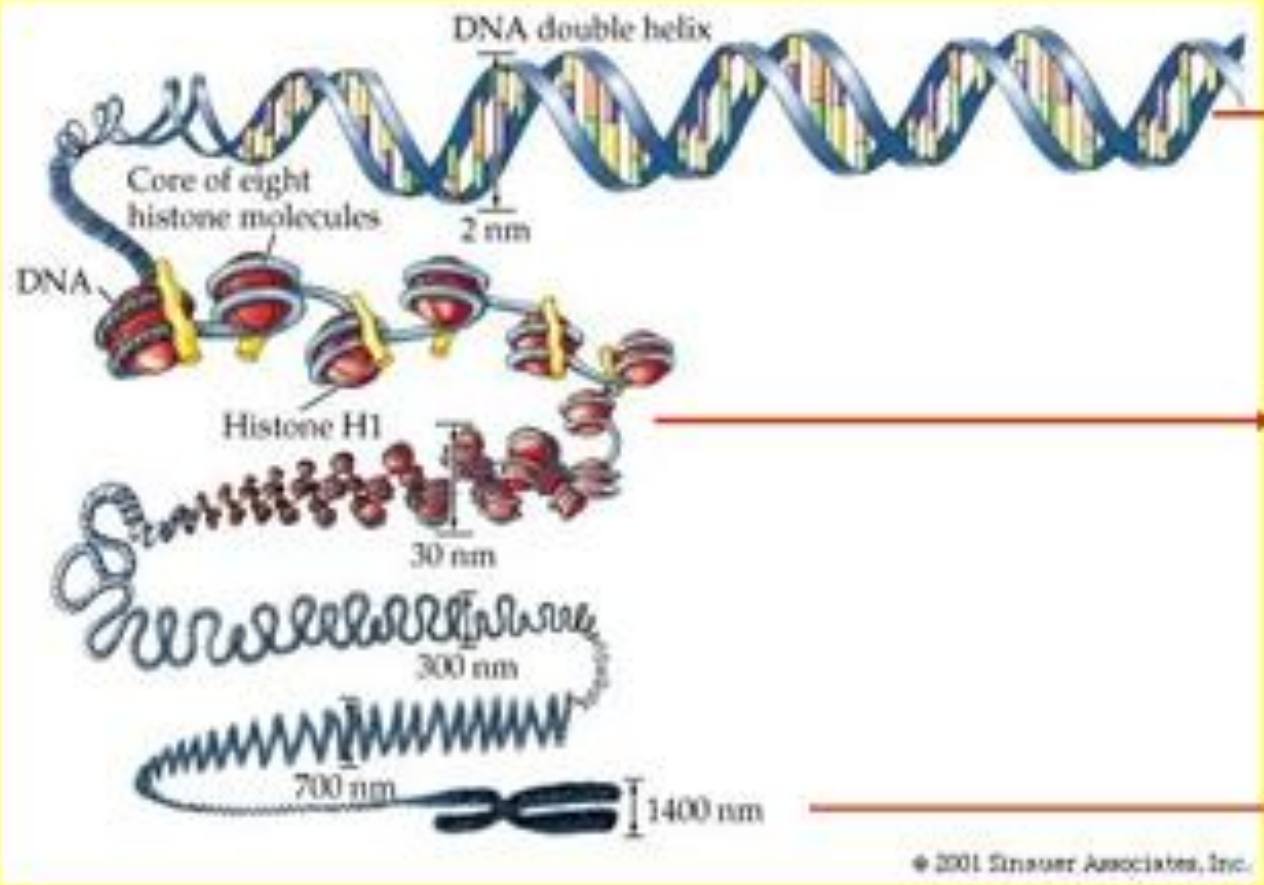


Where Do I Find DNA?

- Chromosomes are in the nucleus of every cell.
- Chromosomes are made up of DNA.
- Genes are pieces of DNA that contain the instructions for building a protein.



Chromosomes are made of DNA and proteins



DNA the "double helix" is a polynucleotide

The DNA winds over proteins (HISTONES) to form chromosomes

The genetic material of the cell is in its chromosomes

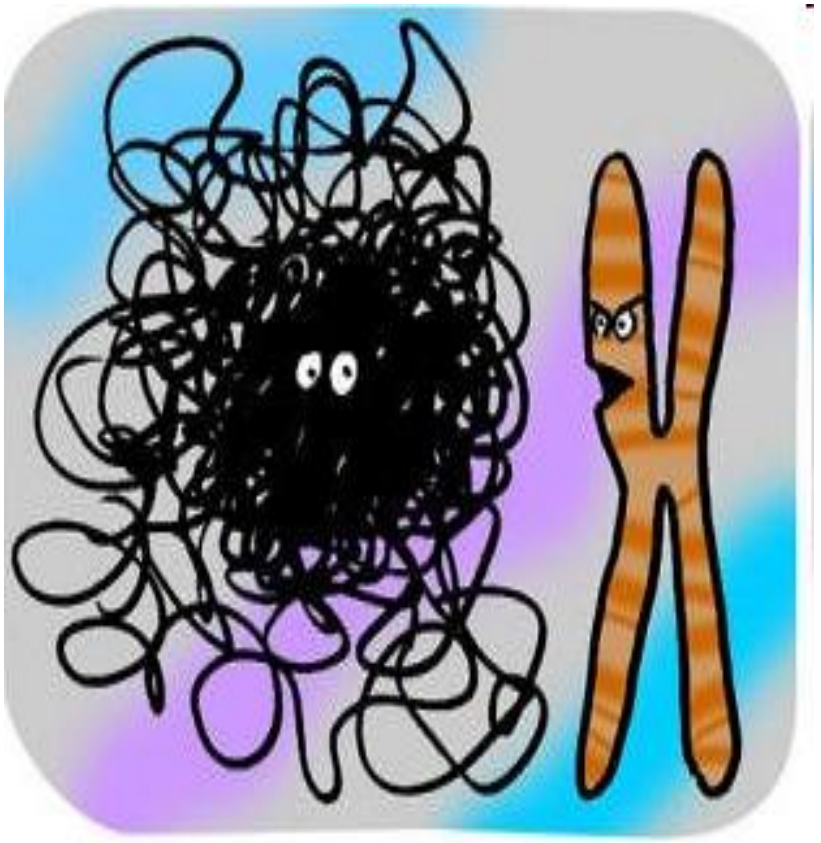
Packing for the move…

When the cell is not dividing…

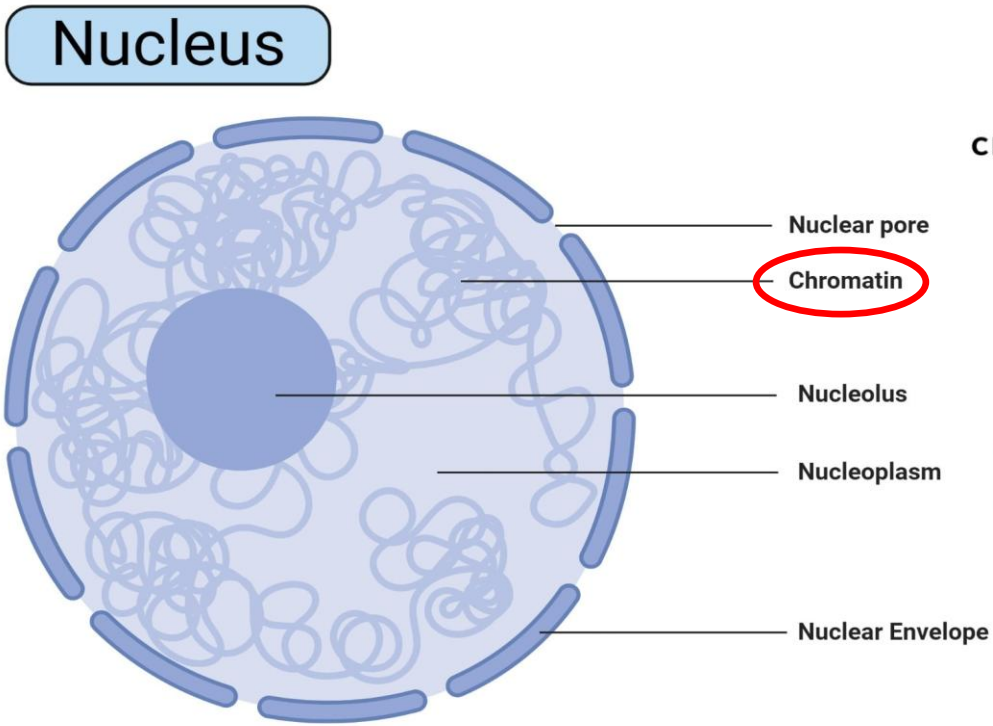
- DNA molecules are in extended, uncondensed form = **chromatin**
- Cell can only replicate and transcribe DNA when it is in the extended state.

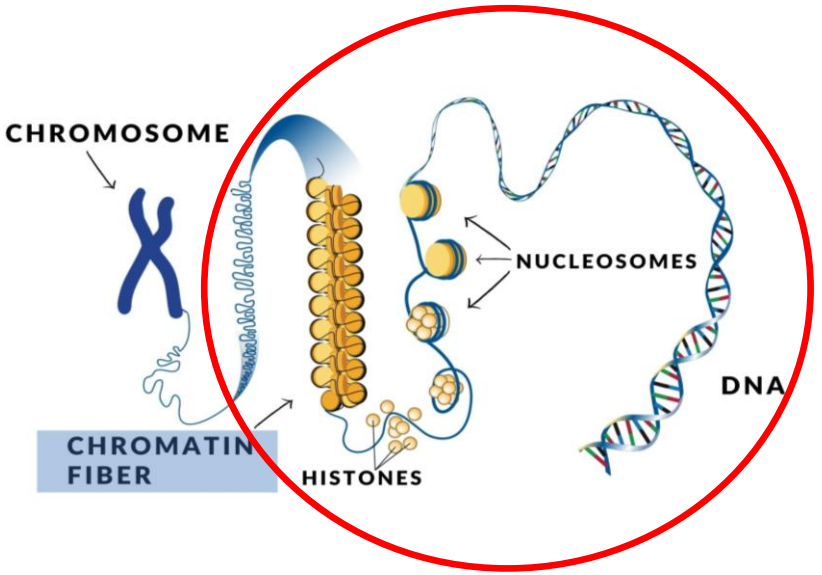
When the cell is preparing for division…

- DNA molecules condense to form **chromosomes** prior to division.
  - each chromosome is a single molecule of DNA
  - easier to sort and organize the replicated DNA into daughter cells



Structure of a nucleus- Focus on chromatin fibres and DNA packaging



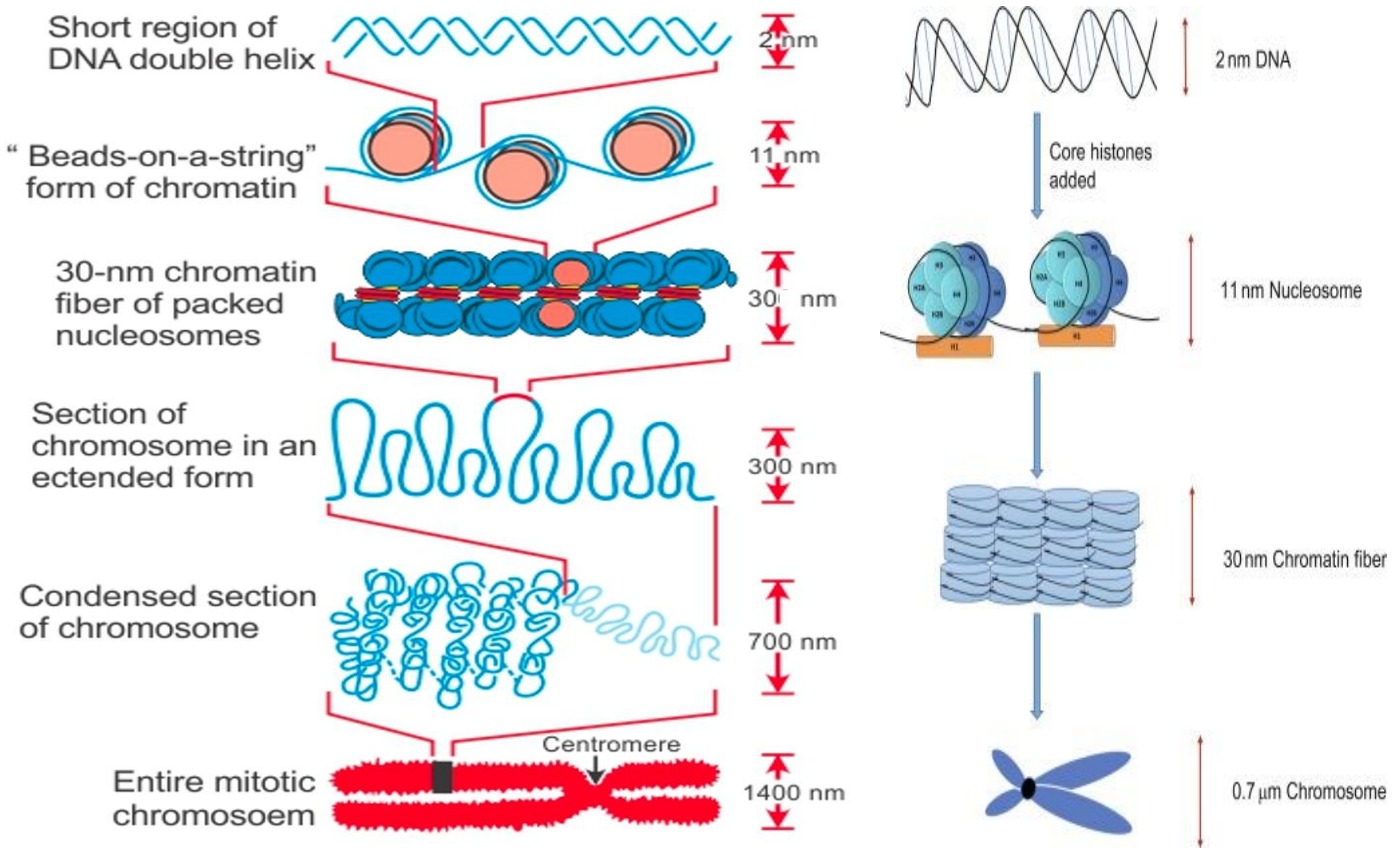


Beads on a string structure

Chromatin fibres: A chromatin fiber is a long thread-like material in the cell nucleus made of DNA wrapped around histones. It is seen as “beads on a string” under electron microscope.

- **Euchromatin:** A loosely packed fiber. Lightly stained. Active regions. It lets the cell read the DNA easily to make proteins.
- **Heterochromatin:** A tightly packed fiber. Dark color stains. It keeps the DNA locked up and **inactive regions.**

Packing of DNA into chromosomes – chromosomes are highly condensed form of DNA during cell division



✓ **DNA packaging is the ordered process of folding a very long DNA double helix (about 2.2 meters long in a human cell) into a tiny nucleus (about 10^{-6} meters wide)**

30nm Chromatin Fiber- exists in the form of solenoid

Solenoid model: In this concept, a linear array of nucleosomes is wound into a solenoid of higher order, a left-handed helix with about six nucleosomes on each turn.

Nucleosomes:

- Beads-like-structure in chromatin fibre , which is made up of four core histone proteins- **H2A, H2B, H3 and H4** in two copies each, around which DNA is wrapped. The core histone proteins exist in octamer (8 copies).
- **146 bp DNA fragment** is wrapped around the core histone proteins. Nucleosomes appear as **"beads-on-a-string" in chromatin fibres**. **"Beads"** refer to nucleosome and **"string"** refers to linear double-stranded DNA between adjacent nucleosomes and **is called linker DNA**.
- Linker DNA is associated with **H1 histone protein**. It basically locks the DNA in place onto the nucleosome.

H1 protein is bound to DNA as it enters each nucleosome core particle. **The average length of linker DNA is approx. 54 bp**. Overall DNA length is shortened by about seven fold by winding around the nucleosome.

Histones:

- Small, positively charged proteins known as histones come in five main varieties: **H1, H2A, H2B, H3, and H4.**
- H1 attaches to 20–22 bp of DNA at the points where the strands enter and exit the octamer**. H2A, H2B, H3, and H4 forms the core of nucleosomes.**
- High concentrations of **the basic amino acids - arginine and lysine** are found in high concentrations in histones.
- These positively charged amino acids provide the **histones a net positive charge,** which makes it easier for them to connect to the negatively charged DNA. **The positively charged histone protein forms ionic bond with negatively charged phosphates of DNA.**
- Even among animals that are distantly related, the amino acid sequences of histones H2A, H2B, H3, AND H4 are remarkably conserved. These amino acid sequences' evolutionary constancy strongly suggests that histones play the same fundamental function in arranging the DNA in chromosomes throughout all eukaryotes

The region of DNA in base pairs that are associated with one nucleosome is 146 bp + 54 bp. This is equal to 200 bp.

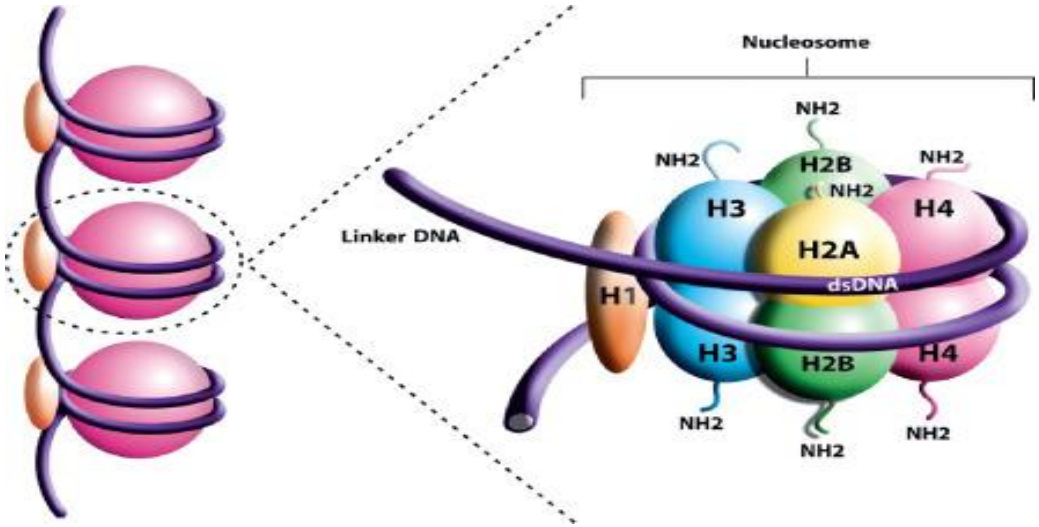
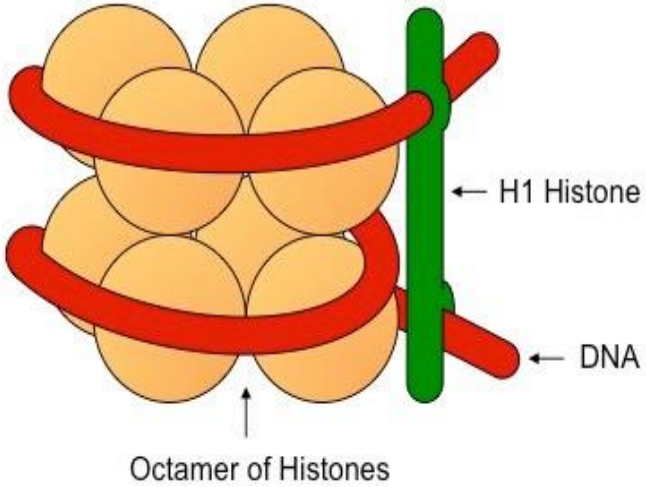
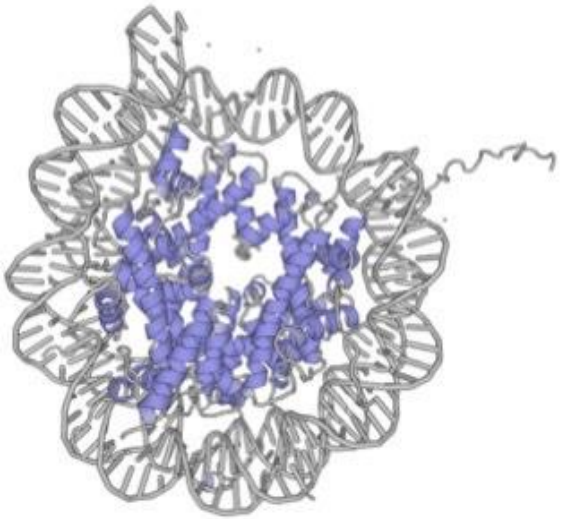


Diagram of a Nucleosome (SIDE VIEW)

Nucleosome Molecule (TOP VIEW)



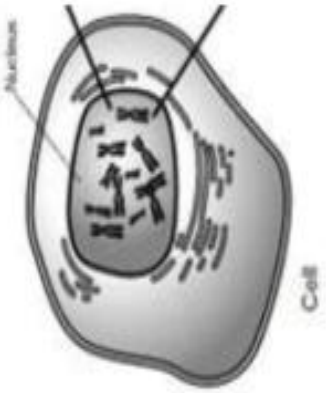


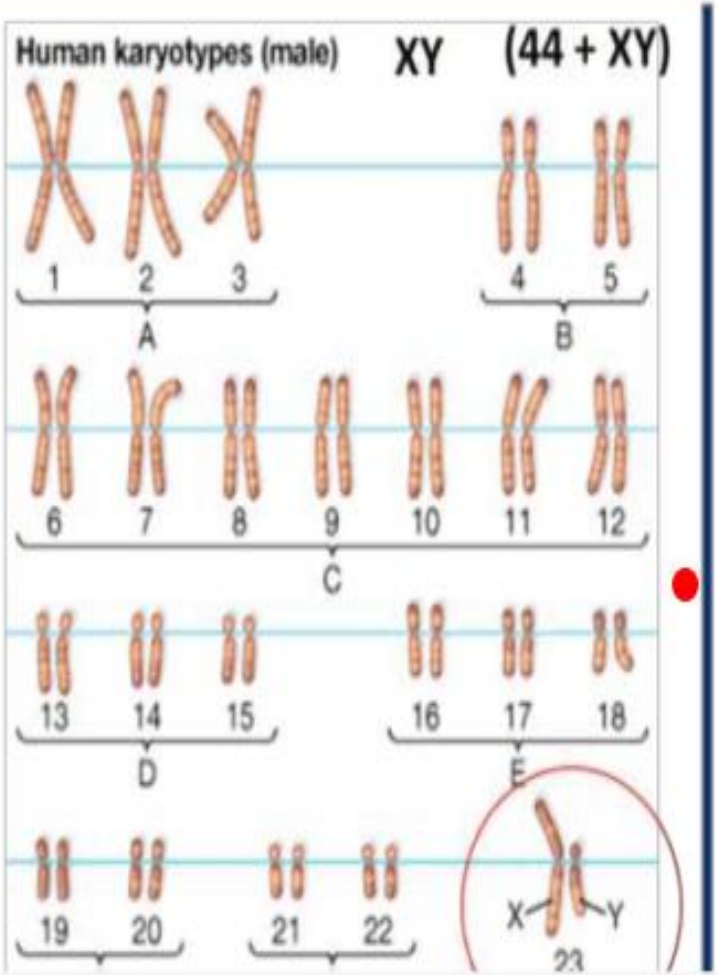
Human Karyotype of male and female (orderly arrangement of chromosomes)

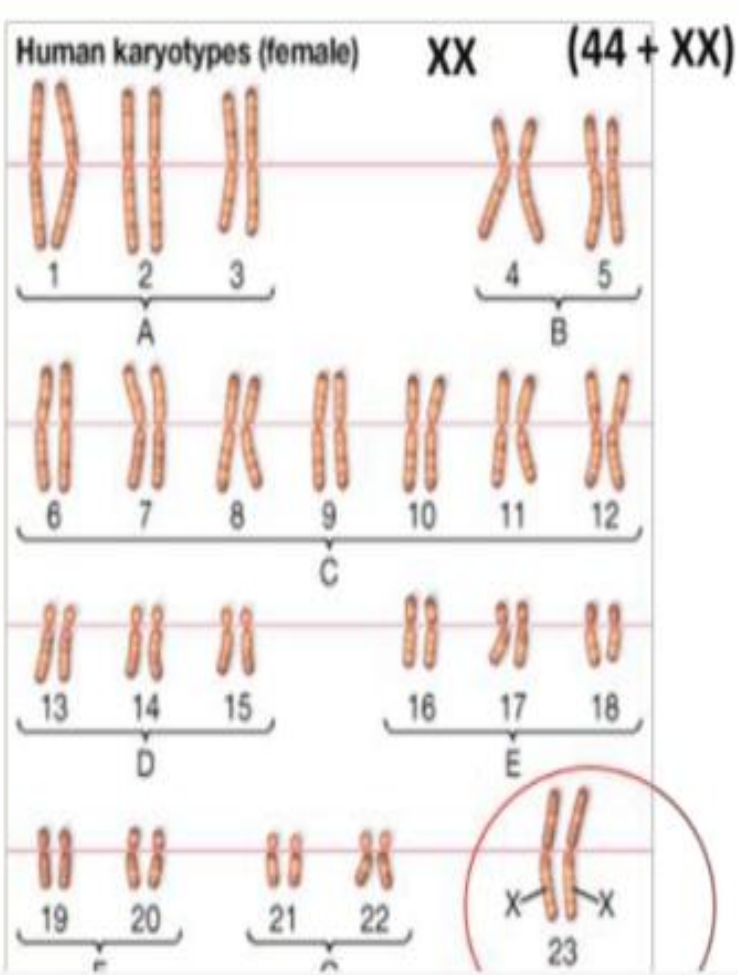
A Karyotype is the map of individual's chromosomes, Homologous pairs of chromosomes of an individual cell. Since every cell in the human body has the same DNA an therefore the same chromosomes, the analysis of one cell is enough.

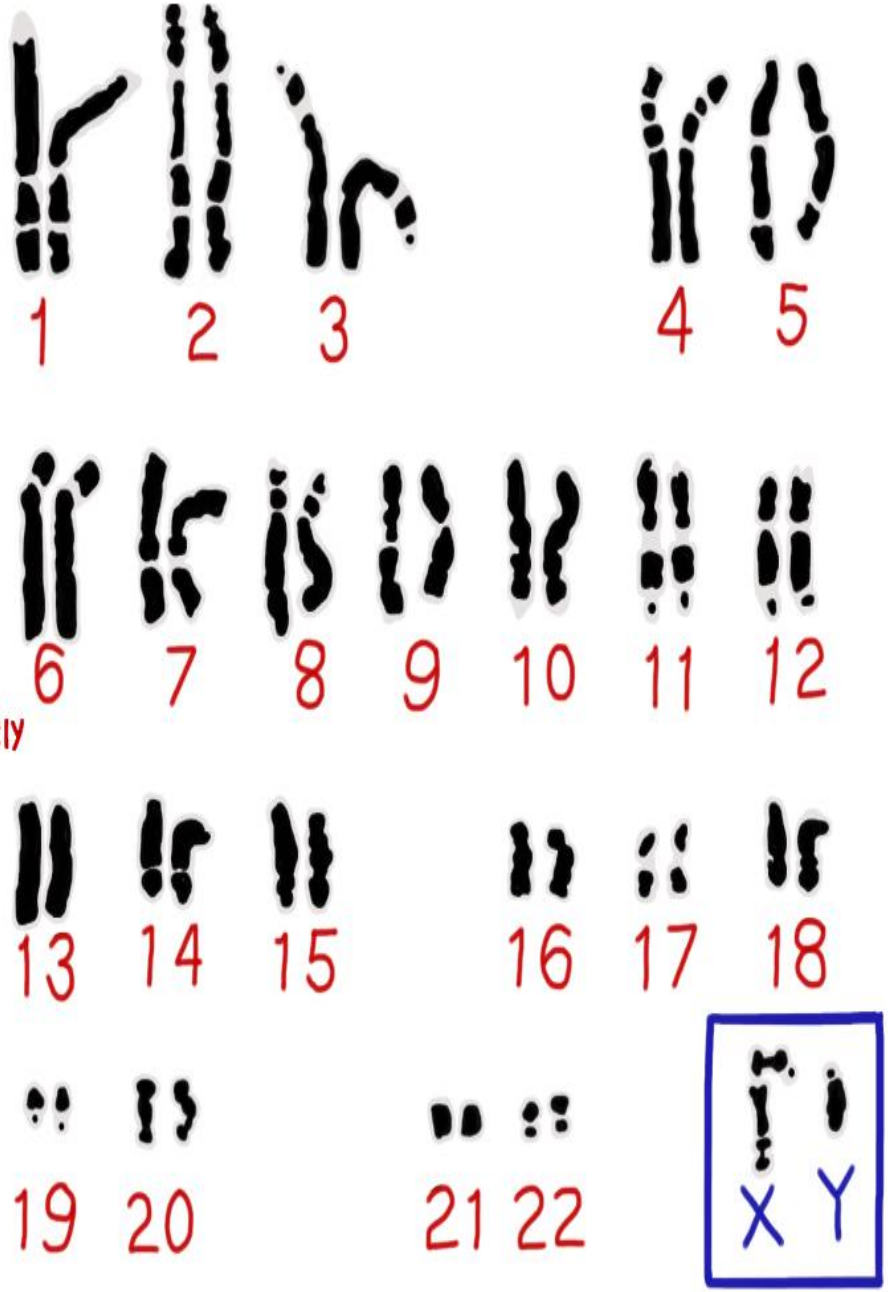
Human has 46 chromosomes, or 23 pairs of chromosomes. Also called , 2n or dipoid

Chromosomes in duplicated forms inside the nucleus







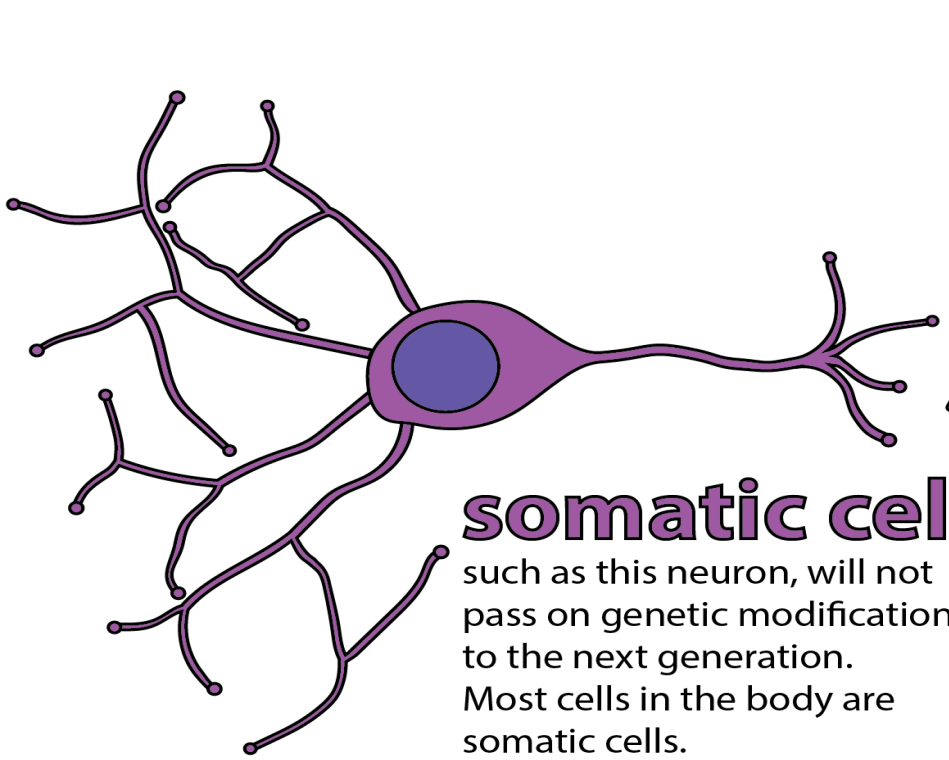


autosome

A chromosome that is not directly involved in determining the sex of an organism.

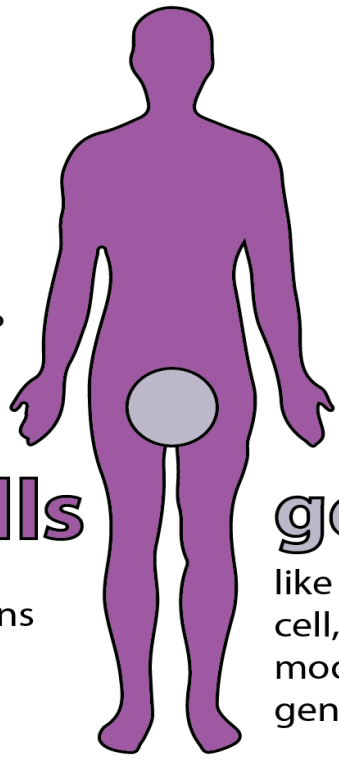
sex chromosome

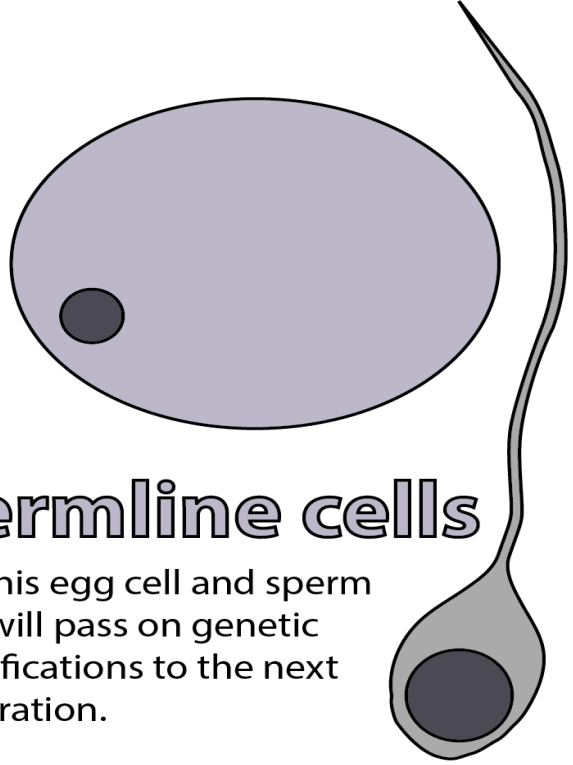
A chromosome that determines whether an organism is male or female.



somatic cells

such as this neuron, will not pass on genetic modifications to the next generation. Most cells in the body are somatic cells.





germline cells

like this egg cell and sperm cell, will pass on genetic modifications to the next generation.

Somatic Cells vs Gametic Cells

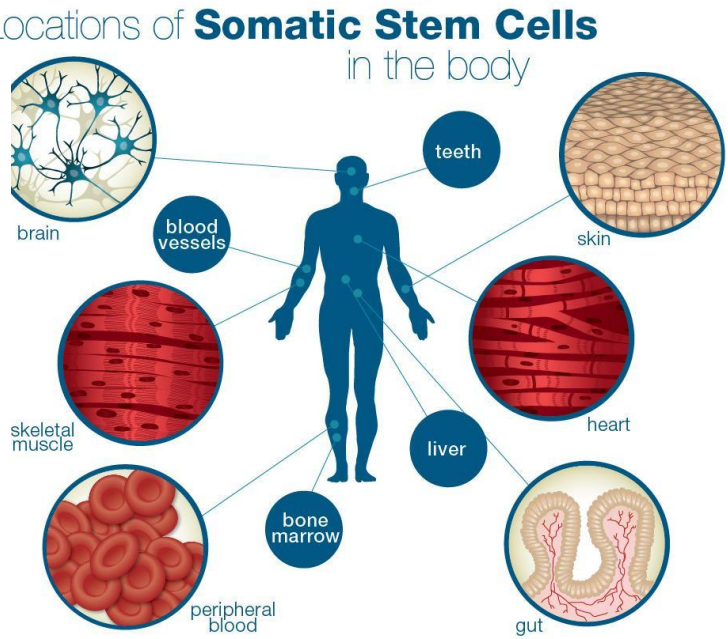
Somatic cells are cells of the body

In humans somatic cells are diploid which means they contain two copies of each chromosome

Human somatic cells have 23 pairs of chromosomes for a total of 46 chromosomes

Gametes are reproductive cells such as sperm and oocytes

Gametes are haploid which means that there is one copy of each chromosome in each gamete

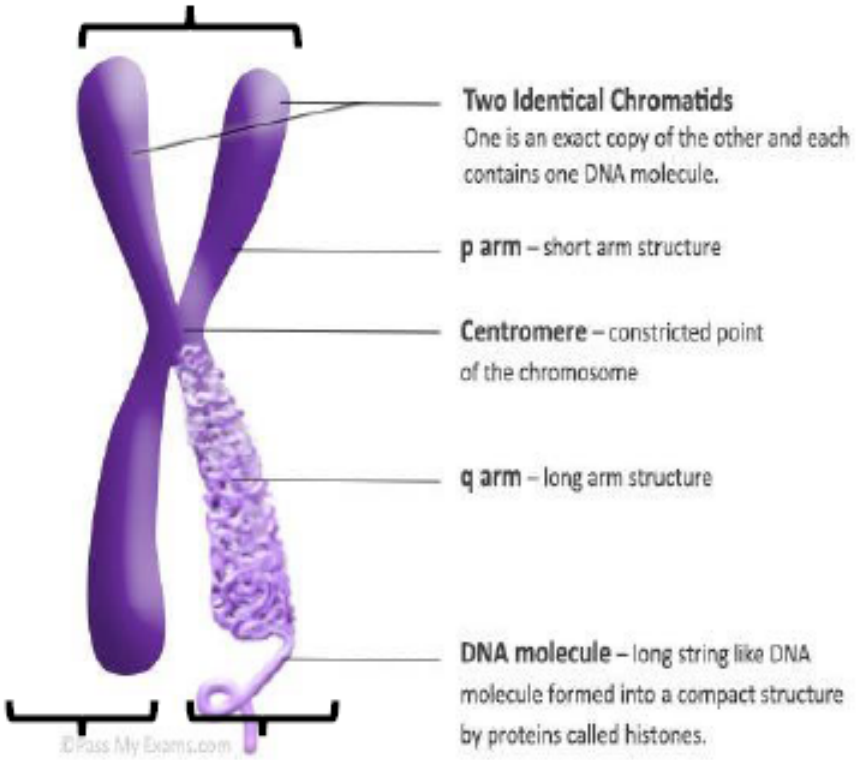


Key words to understand the process of cell divisions

1) Chromosome: A **chromosome** is an organized package of DNA found in the nucleus of the cell. DNA is tightly coiled and package into DNA in chromosome. Different organisms have different numbers of **chromosomes**. Humans have 23 pairs of **chromosomes (so total 46 chromosomes)**--22 pairs of numbered **chromosomes**, called autosomes, and one pair of sex **chromosomes**, X and Y..

2) Chromatid: A chromatid is one copy of a newly copied chromosome which is still joined to the original chromosome by a single centromere. Chromatids are formed after DNA replication. **In other words, chromosome duplicates to form sister chromatids.**

Chromosome

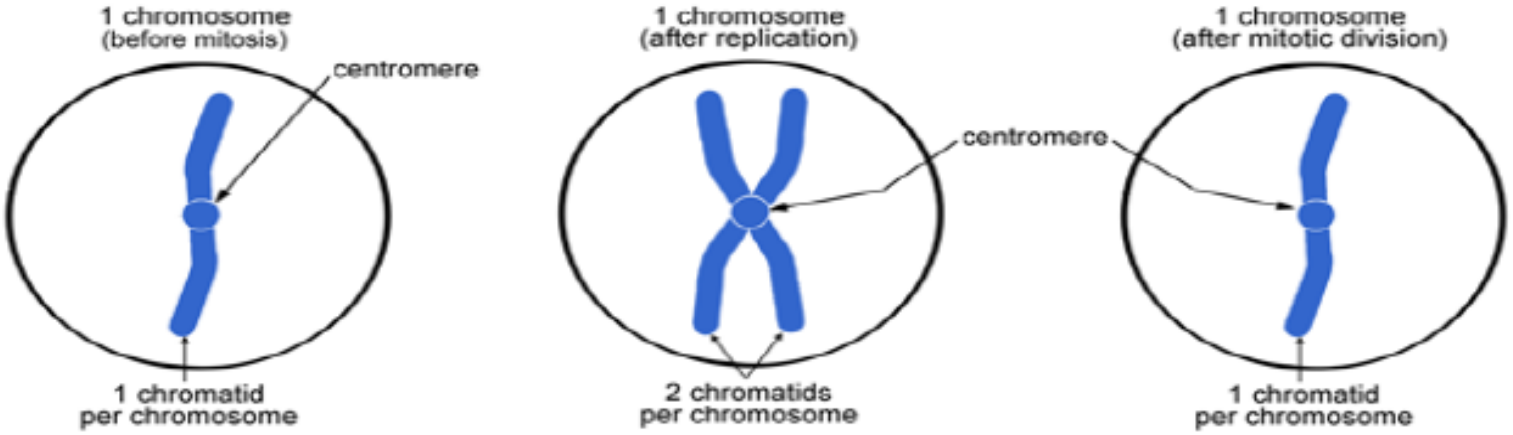


Chromatids: two identical chromatids called sister chromatids,each containing DNA molecule. DNA molecule is replicated.

The concept46 chromosomes----------46 x 2=92 chromatids

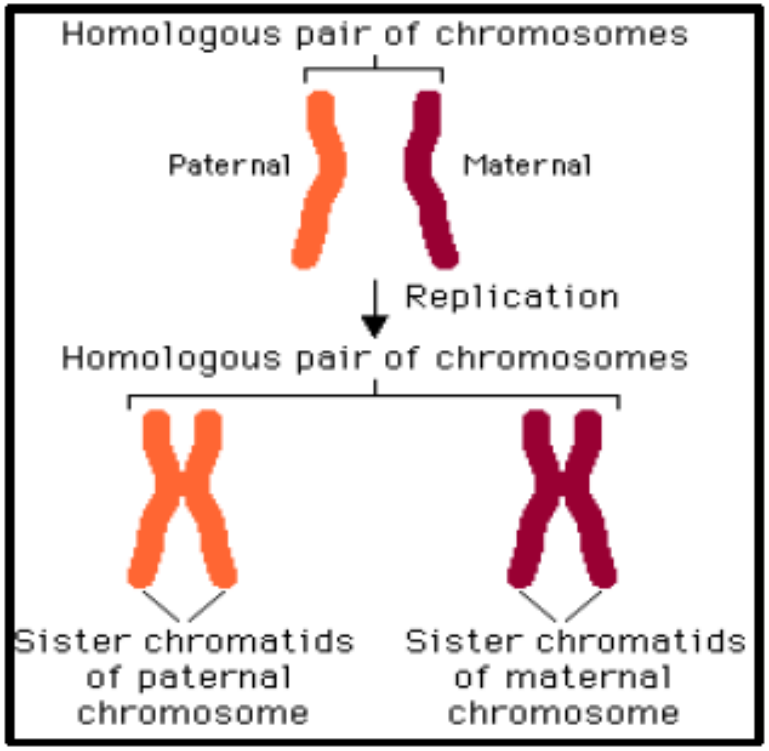
*****The concept from previous slide:

46 chromosomes---------- after DNA replication chromosomes duplicates to form sister chromatids……**46 x 2 = 92 chromatids but 46 chromosomes.**

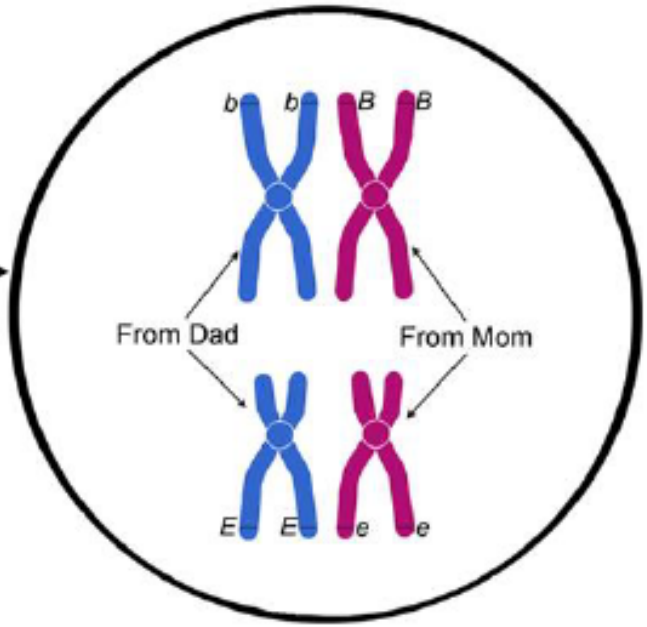


Solve this question?

3) Homologous chromosomes/ homologs/ homologous chromosomes. One pair of chromosomes with the same gene sequence, loci, chromosomal length, and centromere location. In a homologous pair, one chromosome is inherited **from father (paternal)** and other is **inherited from mother (maternal).** The DNA sequences of homologous chromosomes are usually not exactly identical. **Humans have 23 pairs of homologous chromosomes**.



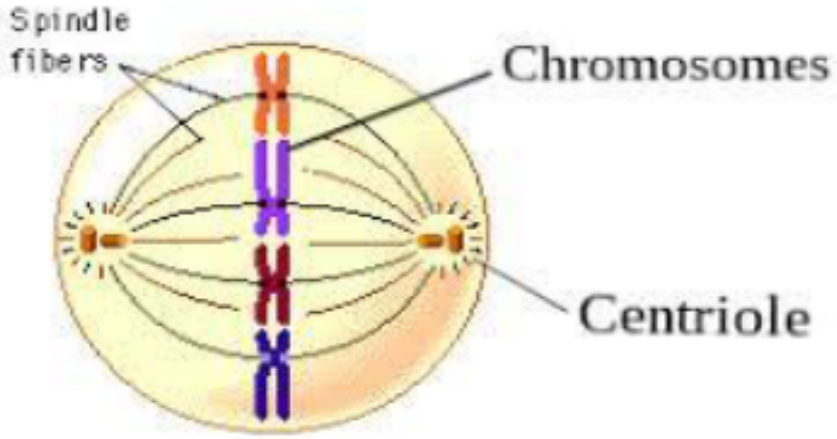
Here, two pairs of homologous chromosomes. One pair blue and pink inherited from dad and mom respectively. **B, b, E and e are gene segments that are aligned together in homologs.**



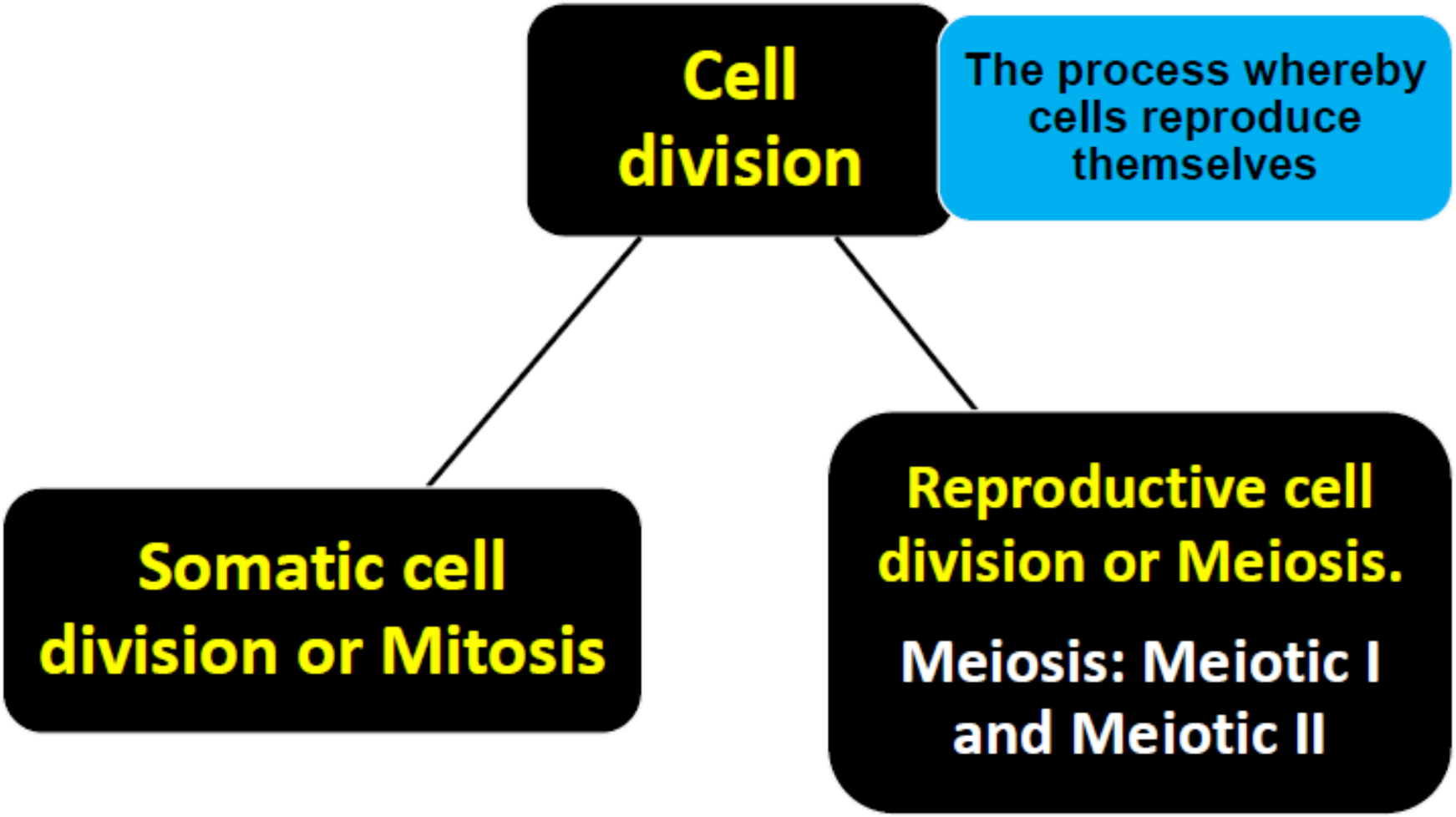
4) Diploid: Diploid is a cell or organism that has paired chromosomes, one from each parent. In humans, all somatic cells other than gametes are **diploid** and have 23 pairs of chromosomes.

5) Haploid: Only single set of chromosomes present in cells. No homologous pairs. For example, Human gametes (egg and sperm cells) contain a single set of chromosomes and are known as haploid.

6) Spindle fibers: They are filaments made up of microtubules that form the mitotic **spindle** in cell division, i.e. mitosis and meiosis. They are chiefly involved in moving and separating the chromosomes into daughter cells during cell division.



After interphase stage of cell cycle, cell divides by two processes either mitosis and meiosis, depending on the cells that is undergoing division.

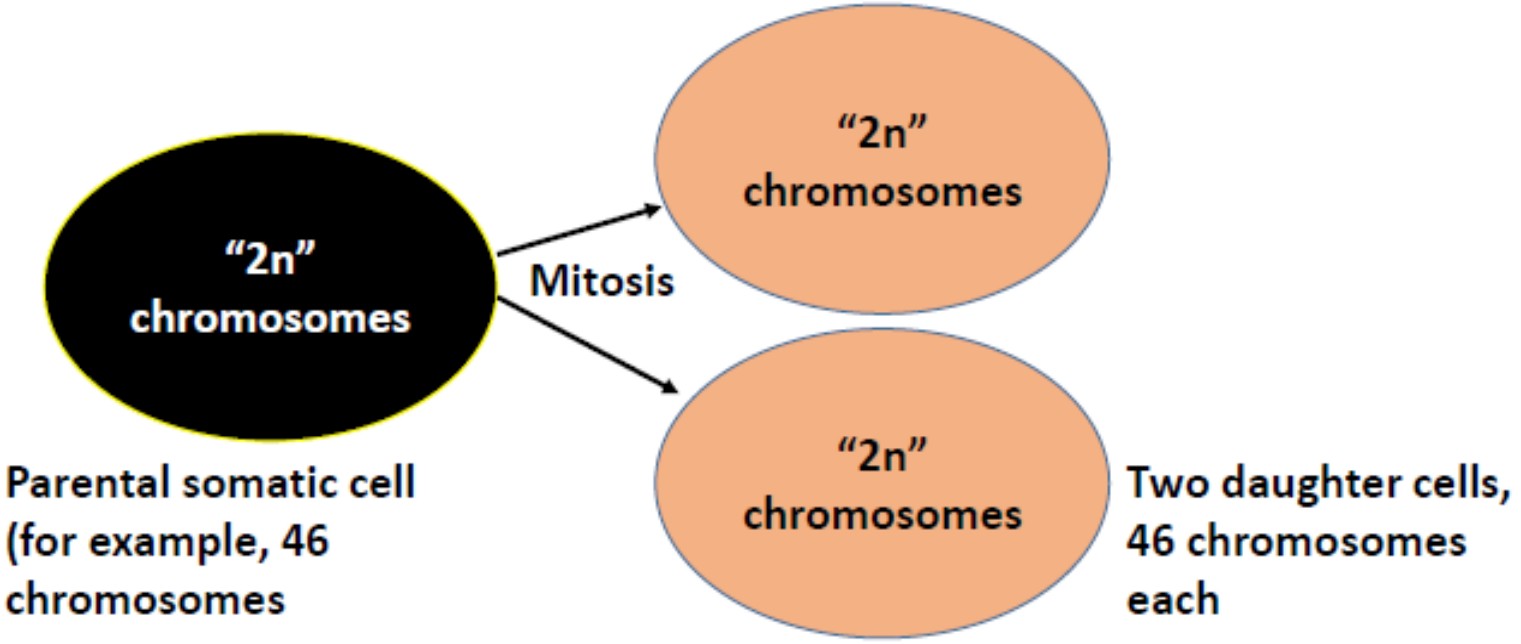


Somatic cell division or Mitosis

1) A cell divides into two identical cells. **(Equational division)**

2) The same genetic material is passed on to the newly formed cells. So after mitosis, each newly formed cell has the same number of chromosomes as the original cell. For example, if the cell has 46 chromosomes, each daughter cell will have 46 chromosomes, **or parental cell diploid state (2n) is maintained in daughter cells**.

3) Significance of mitosis: mitotic cell division maintains replaces dead or injured cells and adds newe ones for tissue growth. For example, skin cells are continually replaced by somatic cell division.



Reproductive cell division or Meiosis

1) **Meiotic cell division occurs** in reproductive cells called **germ cells to produce gametes- sperm and ova. Germ cells are testes in males, and ovaries in females.**

2) Meiotic division **separates** **first pair of homologous chromosomes** (meiotic I: reducing 2n number of chromosomes to "n" chromosomes, and so two daughter cells will half the number of chromosome as its parental cell; meiotic II: mitotic division of the meiotic I daughter cells. So , in meiosis 4 daughter cells are produced with "n" of chromosomes.

After meiosis, **each newly formed cell has the half number of chromosomes as the original cell**. For example, if the cell has 46 chromosomes, each daughter cell will have 23 chromosomes, **or parental cell diploid state (2n) is changed to "n" in daughter cells**.

3) **Significance of meiosis**: the next generation is formed in sexually reproducing organisms. New offsprings are produced.

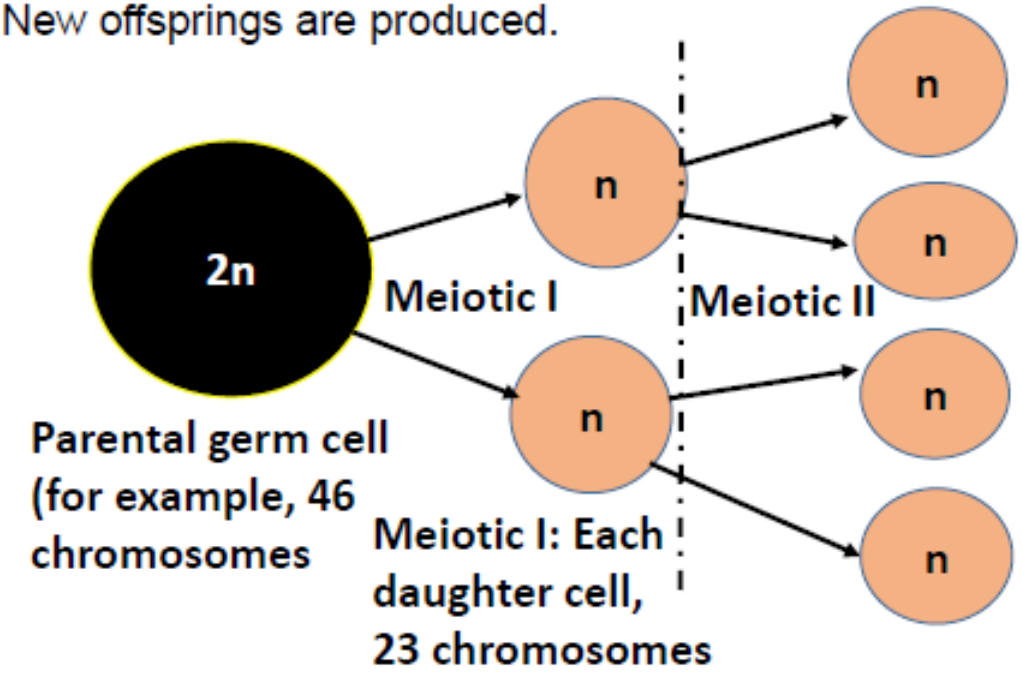


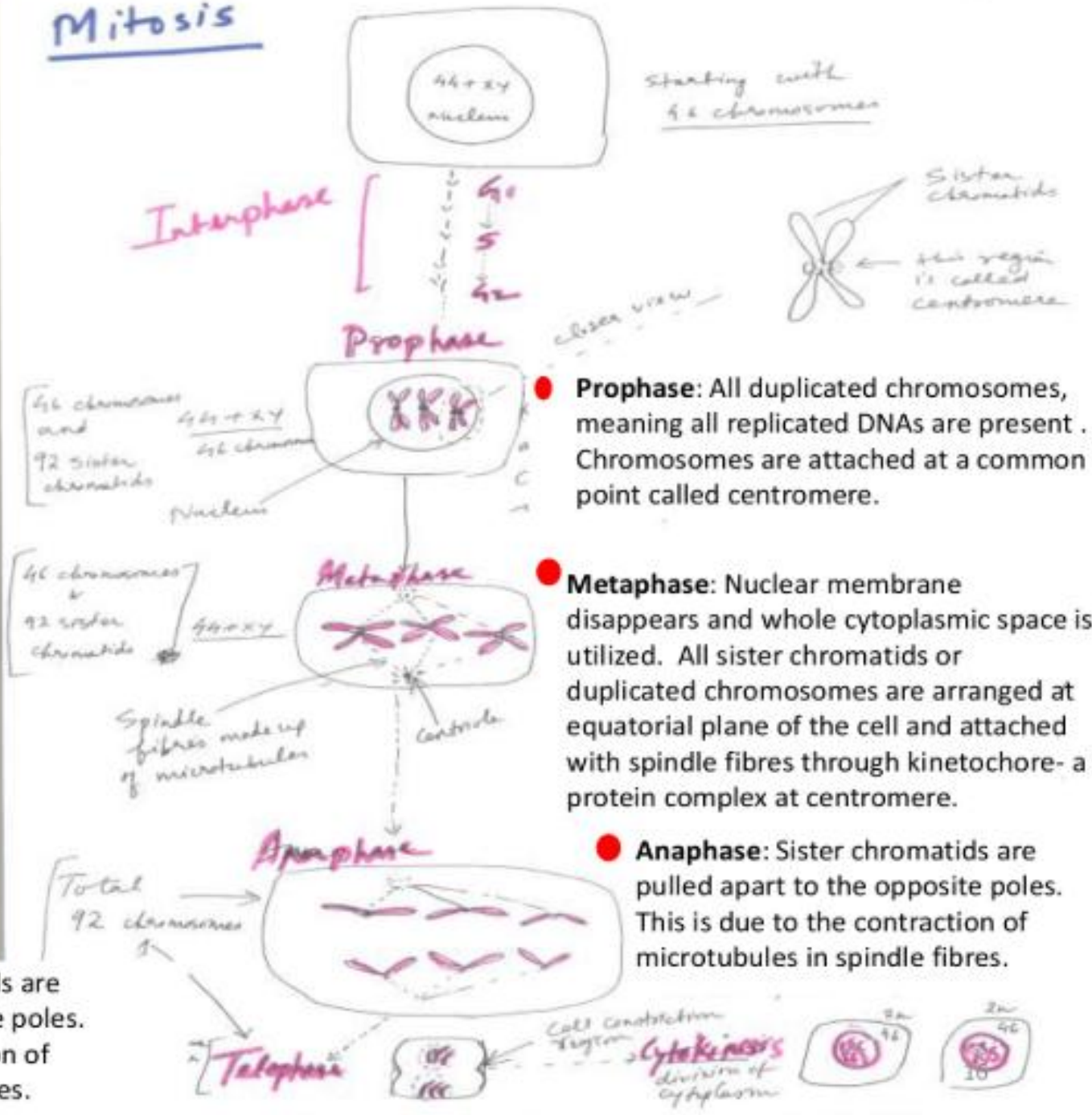
***Figure is explanation of point no. 2

(Reductional division)

Mitosis cell division: different stages

Stages in mitosis: (PMAT)

1. **Interphase**
2. **Prophase-intact nuclear membrane**
3. **Metaphase-no nuclear membrane**
4. **Anaphase-no nuclear membrane**
5. **Telophase-nuclear membrane reappears**
6. **Cytokinesis: division of cytoplasm that leads to two daughter cells with nucleus in each cell**



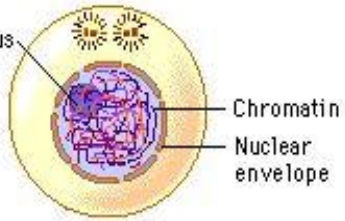
Prophase: All duplicated chromosomes, meaning all replicated DNAs are present . Chromosomes are attached at a common point called centromere.

Metaphase: Nuclear membrane disappears and whole cytoplasmic space is utilized. All sister chromatids or duplicated chromosomes are arranged at equatorial plane of the cell and attached with spindle fibres through kinetochore- a protein complex at centromere.

Anaphase: Sister chromatids are pulled apart to the opposite poles. This is due to the contraction of microtubules in spindle fibres.

Telophase: Sister chromatids are pulled apart to the opposite poles. This is due to the contraction of microtubules in spindle fibres.

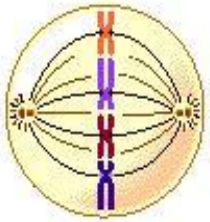
Interphase
The nucleolus and the nuclear envelope are distinct and the chromosomes are in the form of threadlike chromatin.



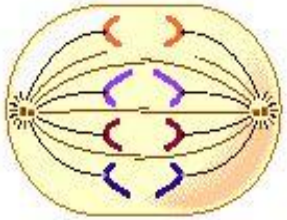
Prophase
The chromosomes appear condensed, and the nuclear envelope is not apparent.



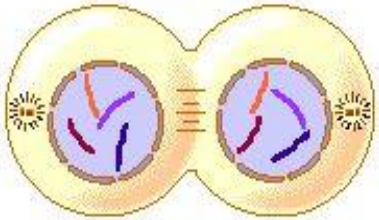
Metaphase
Thick, coiled chromosomes, each with two chromatids, are lined up on the metaphase plate.



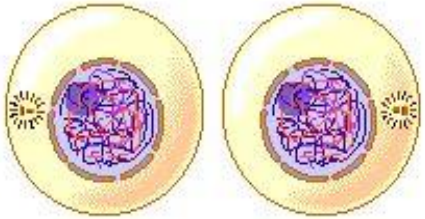
Anaphase
The chromatids of each chromosome have separated and are moving toward the poles.

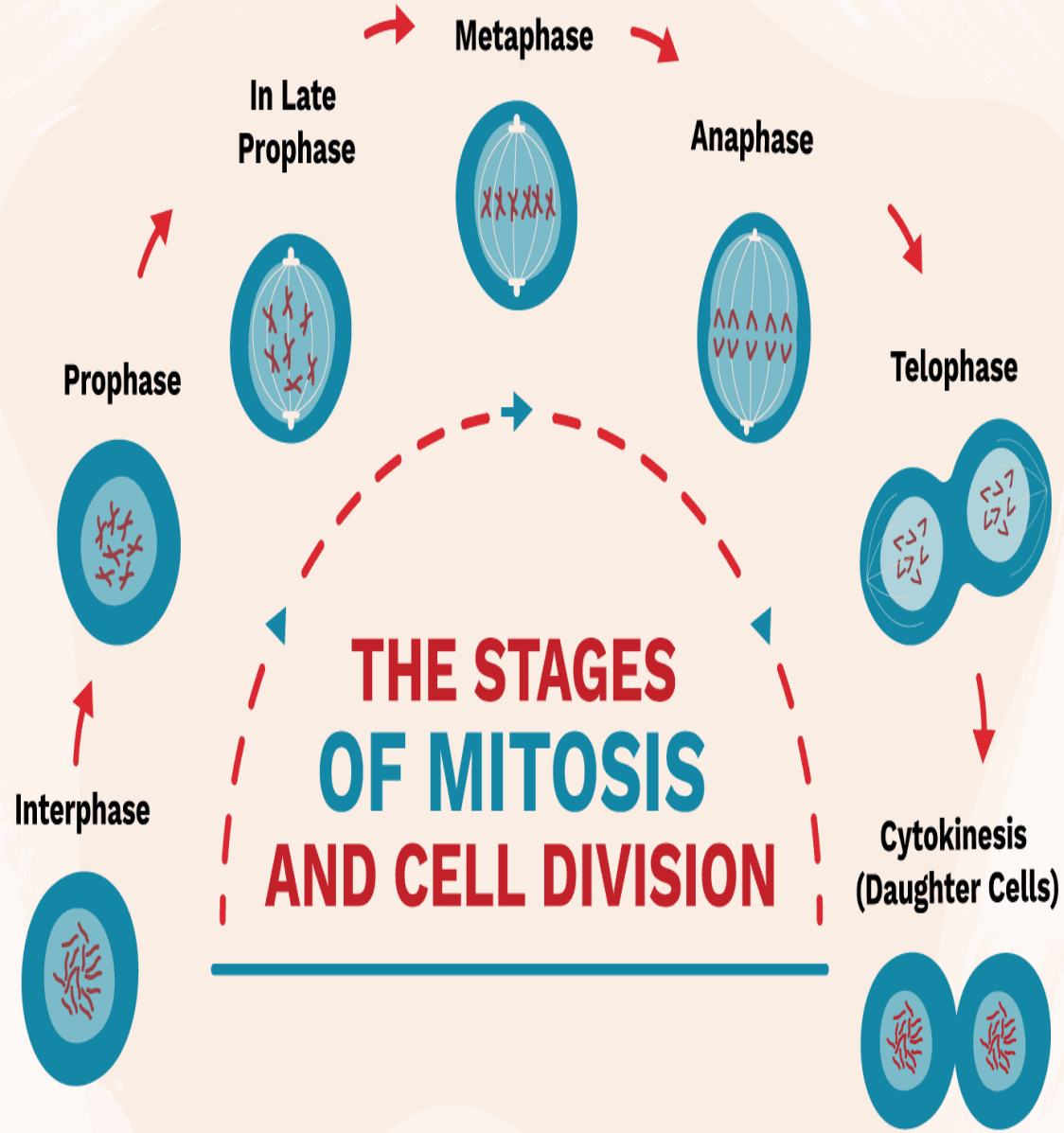


Telophase
The chromosomes are at the poles, and are becoming more diffuse. The nuclear envelope is reforming. The cytoplasm may be dividing.



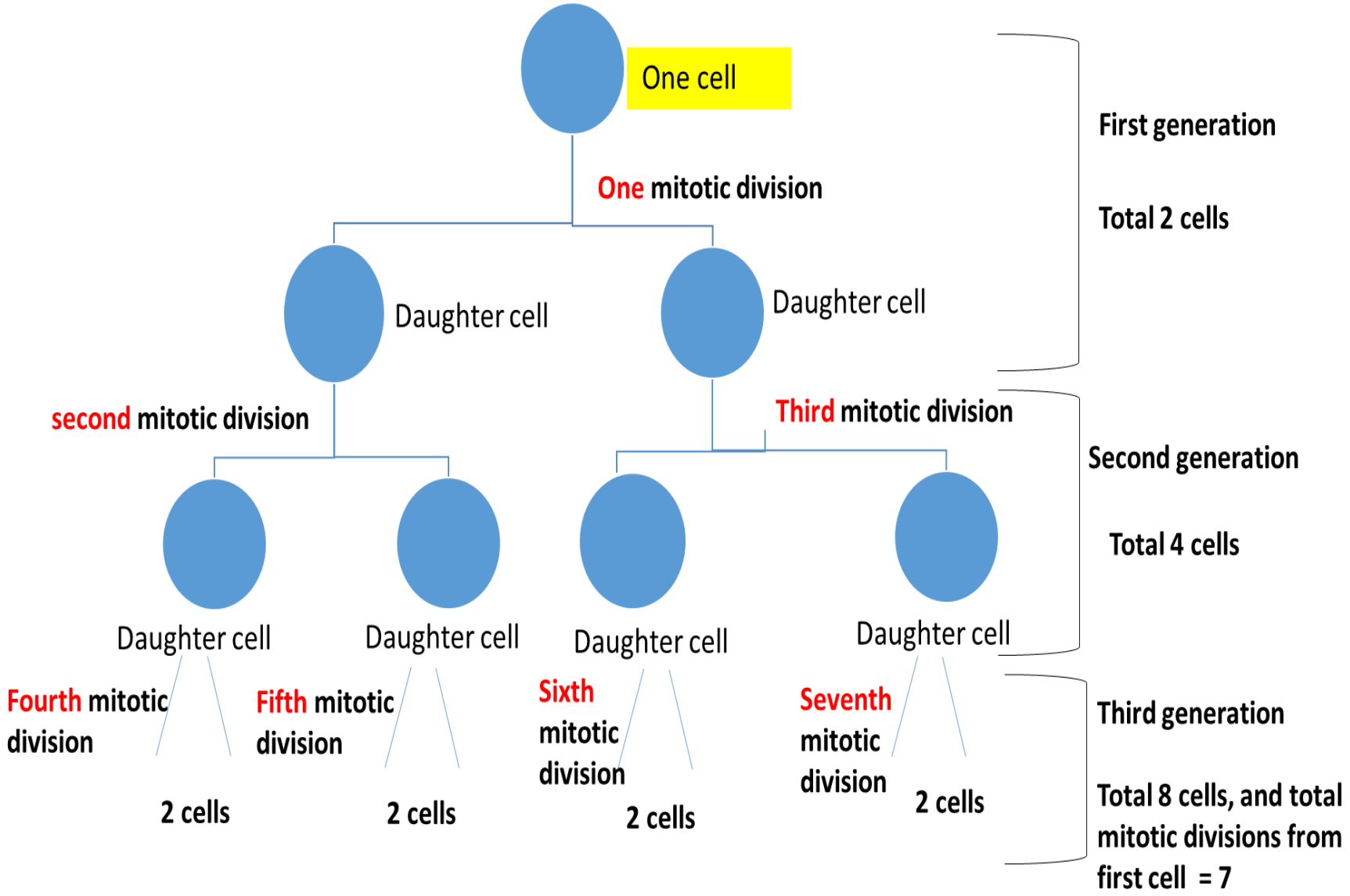
Cytokinesis
Division into two daughter cells is completed.





ThoughtCo.

Cell structure and cell division

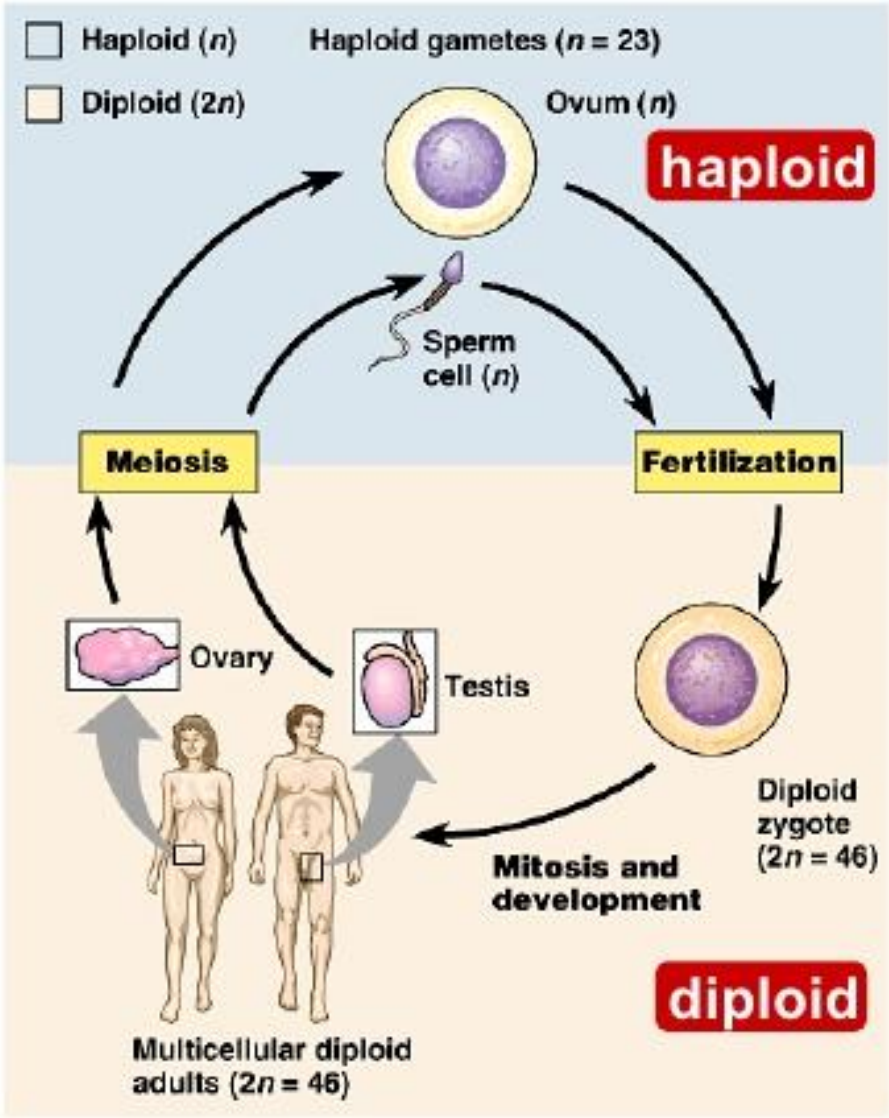


To get 4 cells from one cell, number of mitotic divisions required = 4 – 1= 3 mitotic divisions

Conclusion is, If X is the total number of cells after mitotic divisions, the number of mitotic divisions starting from first cell will be X-1

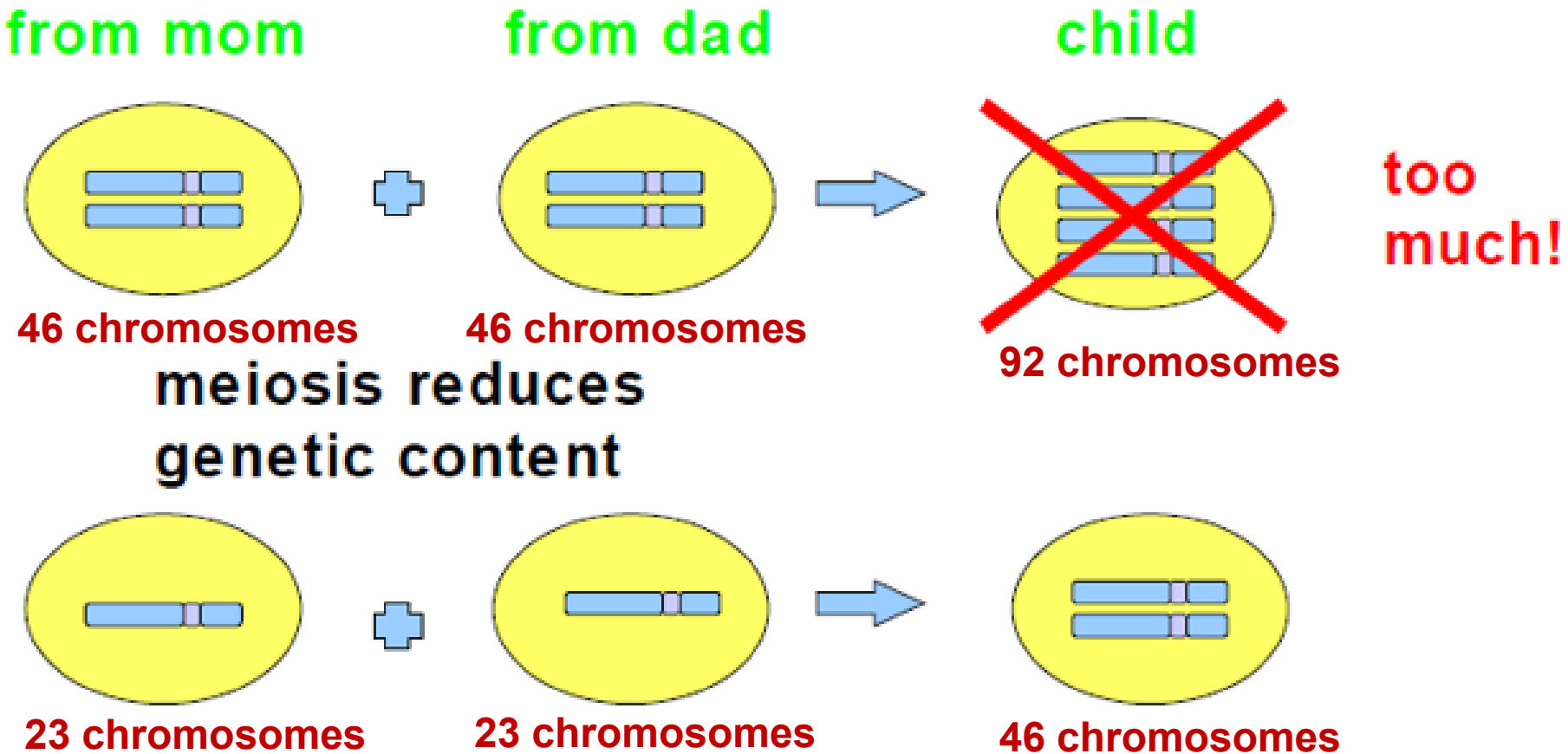
Meiosis: production of gametes

- **Alternating stages**
  - chromosome number must be reduced
    - diploid → haploid
    - 2n → n
      - humans: 46 → 23
    - meiosis reduces chromosome number
    - makes gametes
  - fertilization restores chromosome number
    - haploid → diploid
    - n → 2n



Why form gametes?

- **Sperm and eggs(ova) are gametes**
- Goal: Reduce genetic material by half

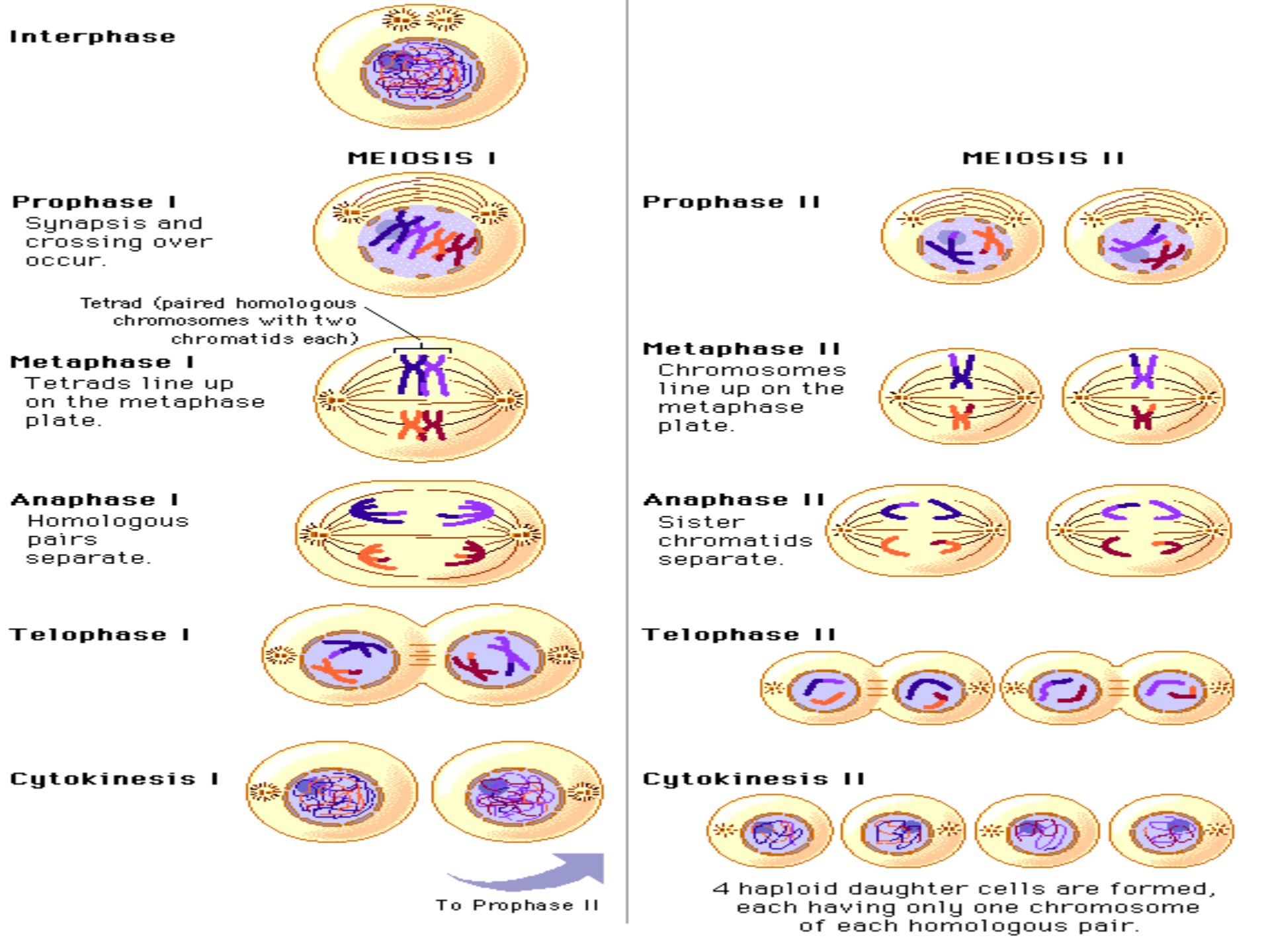




Meiosis is a **process** where a single cell divides twice to produce four cells containing half the original amount of genetic information.

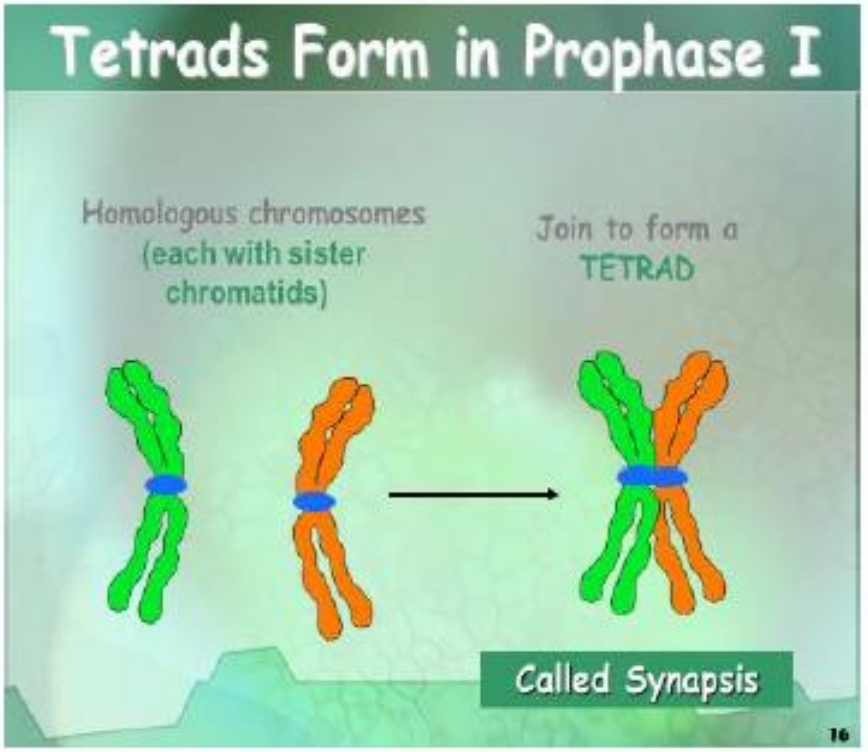
Interphase

Meiosis I	Meiosis II
Prophase I	Prophase II
Metaphase I	Metaphase II
Anaphase I	Anaphase II
Telophase I	Telophase II
Cytokinesis	Cytokinesis


Interphase
MEIOSIS I
Prophase I
Synapsis and crossing over occur.
Tetrad (paired homologous chromosomes with two chromatids each)
Metaphase I
Tetrads line up on the metaphase plate.
Anaphase I
Homologous pairs separate.
Telophase I
Cytokinesis I
To Prophase II
MEIOSIS II
Prophase II
Metaphase II
Chromosomes line up on the metaphase plate.
Anaphase II
Sister chromatids separate.
Telophase II
Cytokinesis II
4 haploid daughter cells are formed, each having only one chromosome of each homologous pair.

Prophase I of meiosis I– important stage for crossing over and exchange of genetic material

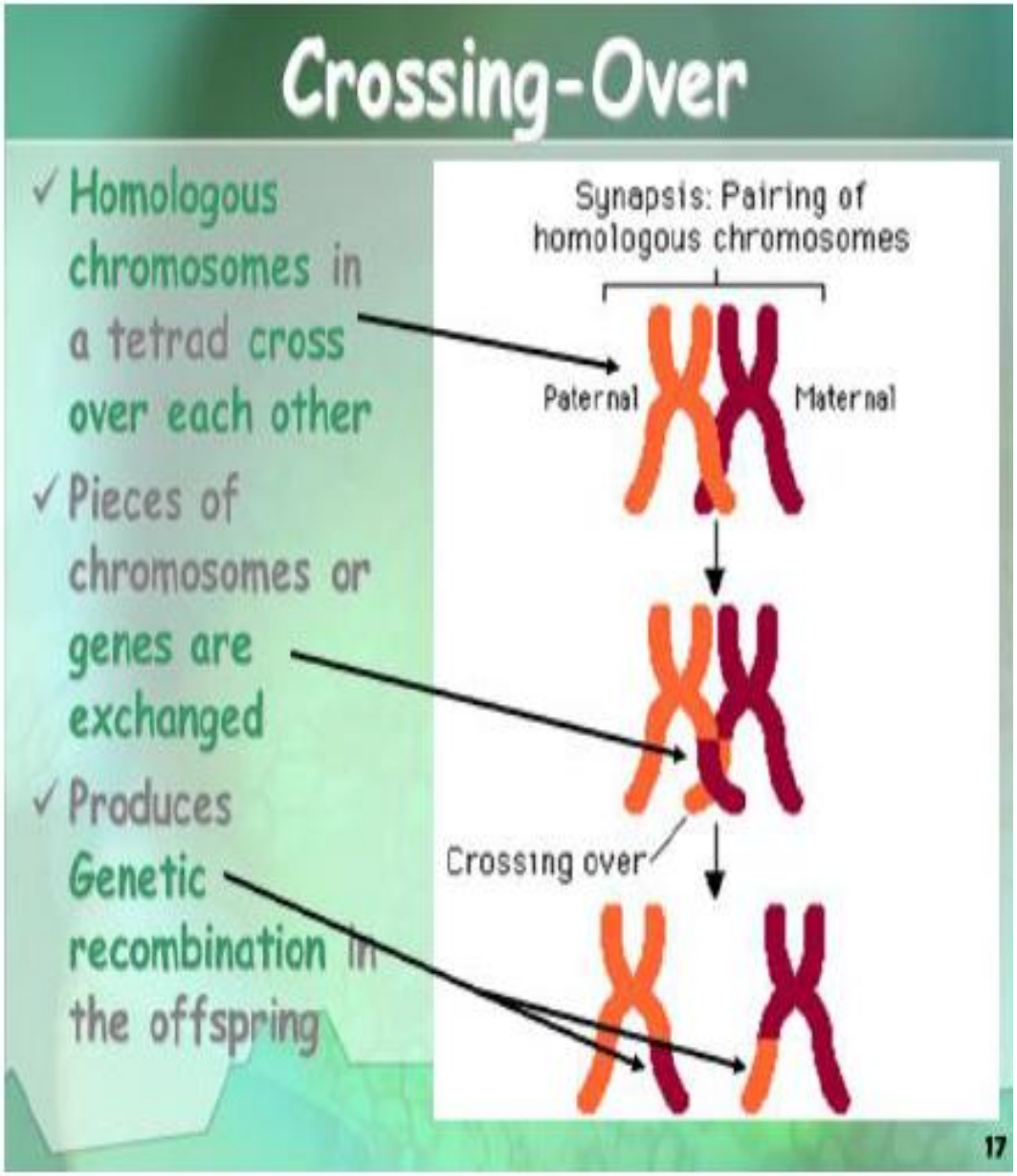


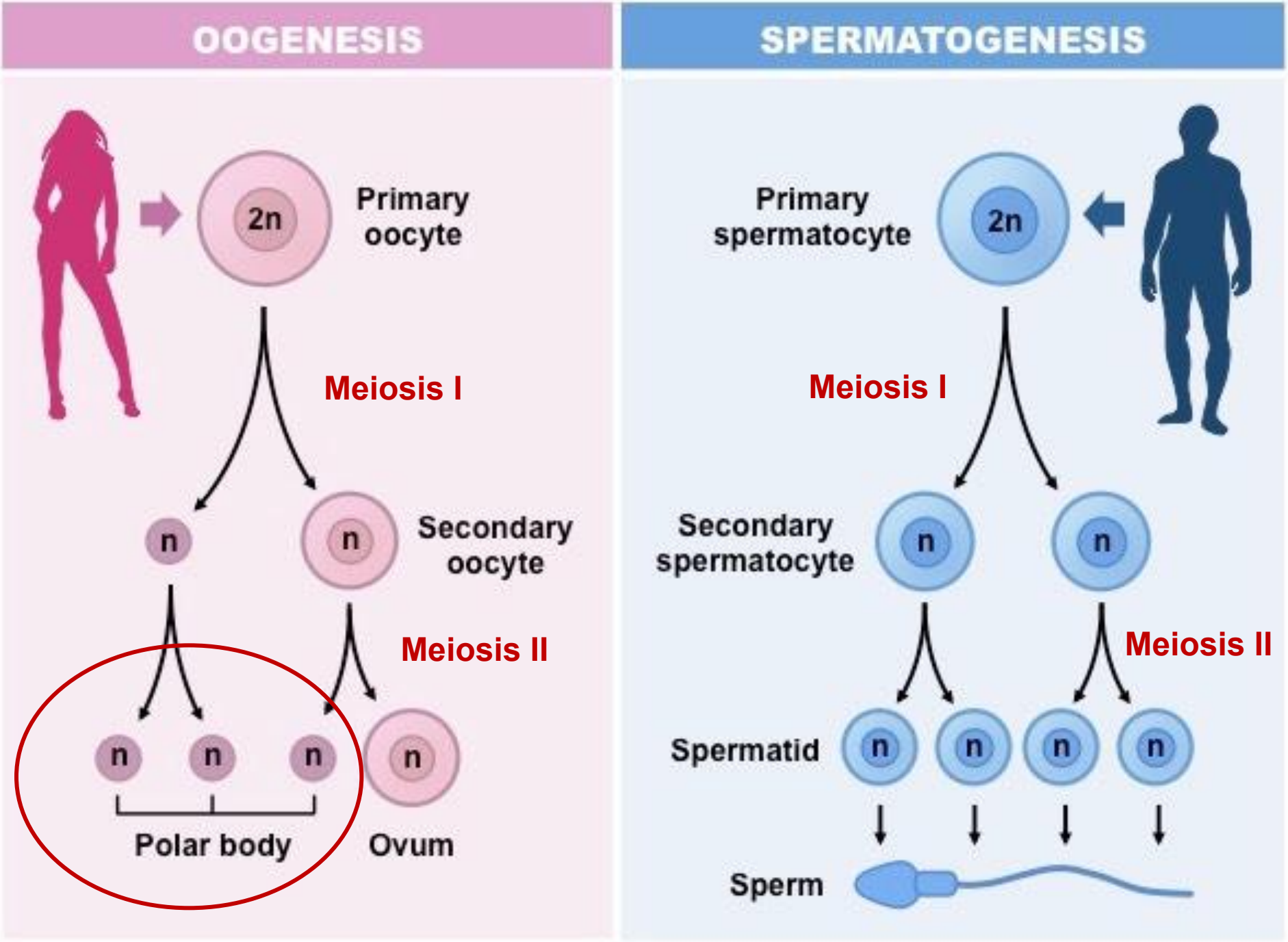


Tetrads formation and crossing over are important processes of meiosis which result in genetic variation among organism. In simple words, there is exchange of genetic material between "mom and dad" that results in offspring "with mom and dad characters".

1) Prophase I is the longest phase of meiosis and divided into 5 stages- leptotene, zygotene, pachytene, diplotene, diakinesis. Zygotene stage is characterized by tetrad formation (Synapsis). Pachytene is characterized by crossing over (exchange of genes) between non sister chromatids.

2) The point at which crossing over takes place is called chiasmata. Exchange of genetic segments produces genetic recombination in the offspring, and brings variation in traits.




OOGENESIS
SPERMATOGENESIS
2n
Primary oocyte
Meiosis I
n
n
Secondary oocyte
Meiosis II
n
n
n
n
Polar body
Ovum
Primary spermatocyte
2n
Meiosis I
Secondary spermatocyte
n
n
Meiosis II
Spermatid
n
n
n
n
Sperm

Chromosomes and chromatids numbers in cell divisions

Phase (Mitosis)	# Chromosomes	# Chromatids
Prophase	46	92
Metaphase	46	92
Anaphase	92	92
Telophase	92	92
End of Mitosis (separated cells)	46	46
Phase (Meiosis I)	# Chromosomes	# Chromatids
Prophase I	46	92
Metaphase I	46	92
Anaphase I	46	92
Telophase I	46	92
End of Meiosis I (separated cells)	23	46
Phase (Meiosis II)	# Chromosomes	# Chromatids
Prophase II	23	46
Metaphase II	23	46
Anaphase II	46	46
Telophase II	46	46
End of Meiosis II (separated cells)	23	23



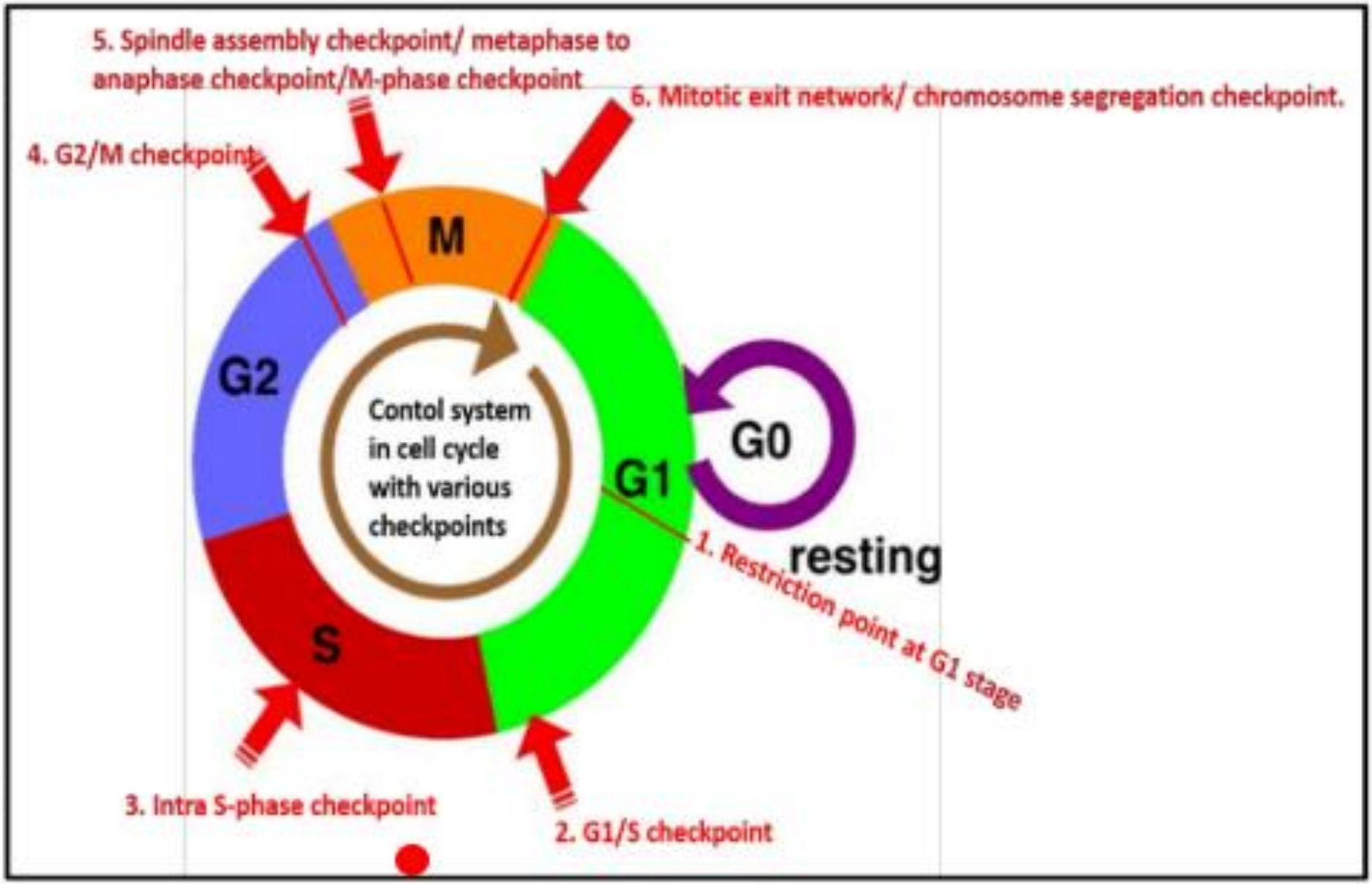
mitosis

1. All organism
2. Asexual
3. Genetically - identical cells
4. No crossing over occurs
5. No pairing of homologous chromosome
6. One cell division occurs
7. 2 diploid cells are produced

meiosis

1. Reproductive cells of humans, plants, animals & fungi
2. Sexual
3. Genetically - Different cells
4. Crossing over occurs
5. Pairing of homologous chromosome
6. Two cell division occurs
7. Four haploid cells are produced

Cell cycle control and different checkpoints



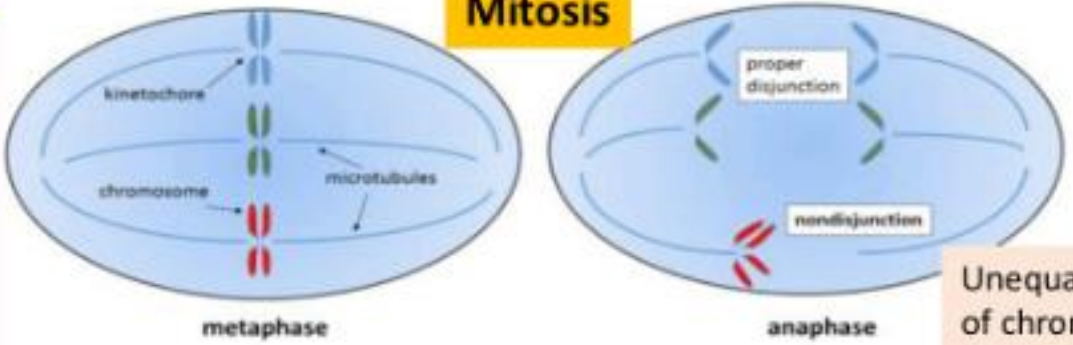
Cell cycle checkpoints/control points	What surveillance /roles these checkpoints do?
1) Restriction point at G1 stage.	a) This is a point of no return at which cell is committed to divide or die by becoming unresponsive to any more external signals. If the cell has improper size or other defects it **enters the temporary stage of resting called G_0 phase**.
2) **G1/S checkpoint**	a) Keeps an eye over DNA damages and ensures that the everything is ready for DNA synthesis. b) Ensures the **proper cell size and availability of nutrients required** for **further stage of the cell cycle**. If any of these are not adequate, there will be **cell arrest and cell go for apoptosis** (cell death).
3) **Intra S-phase checkpoint**	a) This occurs at the middle of the S-phase which is the phase of synthesis of DNA. **b) keeps an eye over proper DNA replication**, any damages or lesions in DNA will stall replication forks during replication process and so on

4) G2/M checkpoint	a) **If DNA synthesis is not completed, cells do not enter** into mitotic phase and goes for apoptosis. b) checks for the mitotic machineries that are required for the next stages. Cell ensures that all the materials are available for carry out division through next stage as well as genome integrity is also maintained
5) Spindle assembly checkpoint or metaphase to anaphase checkpoint	a) This checkpoint **works between metaphase to anaphase stage of M-phase. .** **b) Improper assembly of mitotic spindle fibres prevents the initiation of anaphase.** When spindle fails to attach to the kinetochore of one sister chromatid, the checkpoint do not allow it to enter the anaphase.
6) Mitotic exit network or chromosome segregation checkpoint.	a) **Proper segregation of daughter chromosomes during telophase** and **cytokinesis** is monitored by this checkpoint.

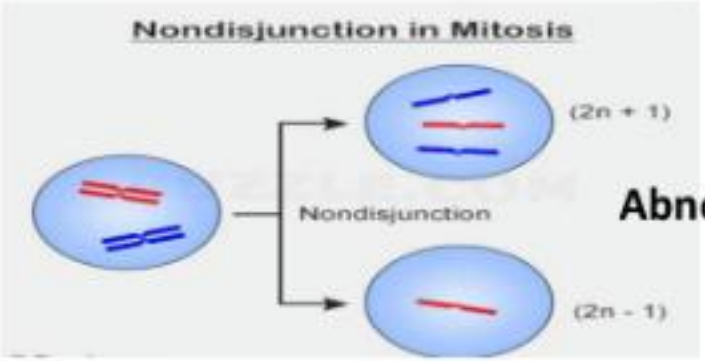
Non disjunction:

Failure of Spindle-assembly checkpoint or metaphase to anaphase transition causing unequal distribution of chromosomes in daughter cells.

Mitosis



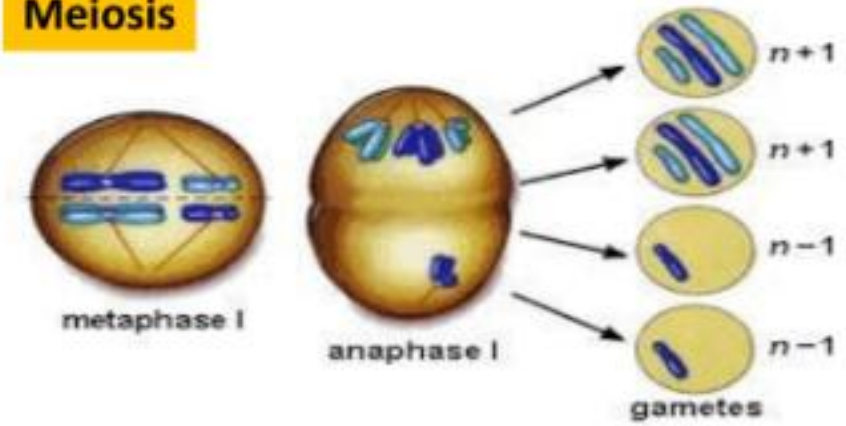
Unequal distribution of chromosome during metaphase to anaphase transition: failure of spindle – assembly checkpoint



Abnormal daughter cells

Meiosis

Why "abnormal" because normal chromosome number is not maintained?

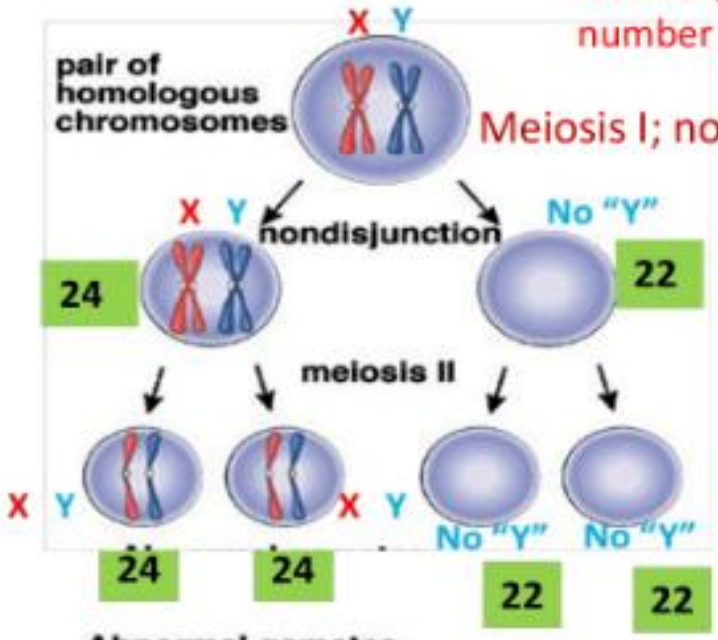


Abnormal gametes

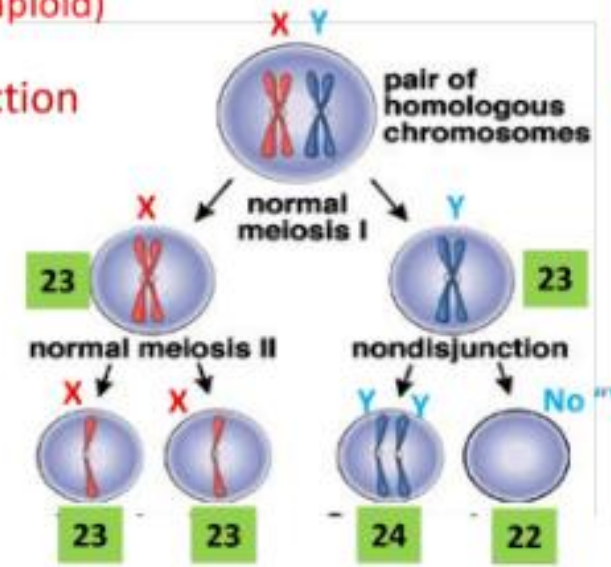
Non-disjunction can happen either in meiosis I … metaphase to anaphase transition or meiosis II during metaphase to anaphase transition

Taking example here the separation of X and Y homologous pair.

Normal gamete chromosome number is : 23 (haploid)



Abnormal gametes: two gametes have one extra and two gametes have one less chromosome than normal.



Gametes have usual number of chromosomes.

One gamete has one extra and the other has one less chromosome.

Meiosis II; nondisjunction in **one of the daughter cells.**

"Abnormal Sperms"..... **22 + XY** (24 chromosome) X Normal ovum..... **22 + X**

Fertilization ↓

44 + XXY **Klinefelter's syndrome**

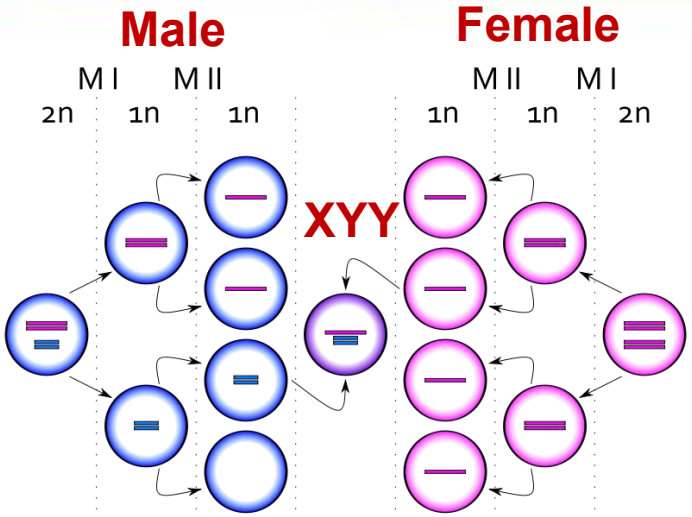
"Abnormal Sperms"..... **22 + O** (22 chromosome) X Normal ovum..... **22 + X**

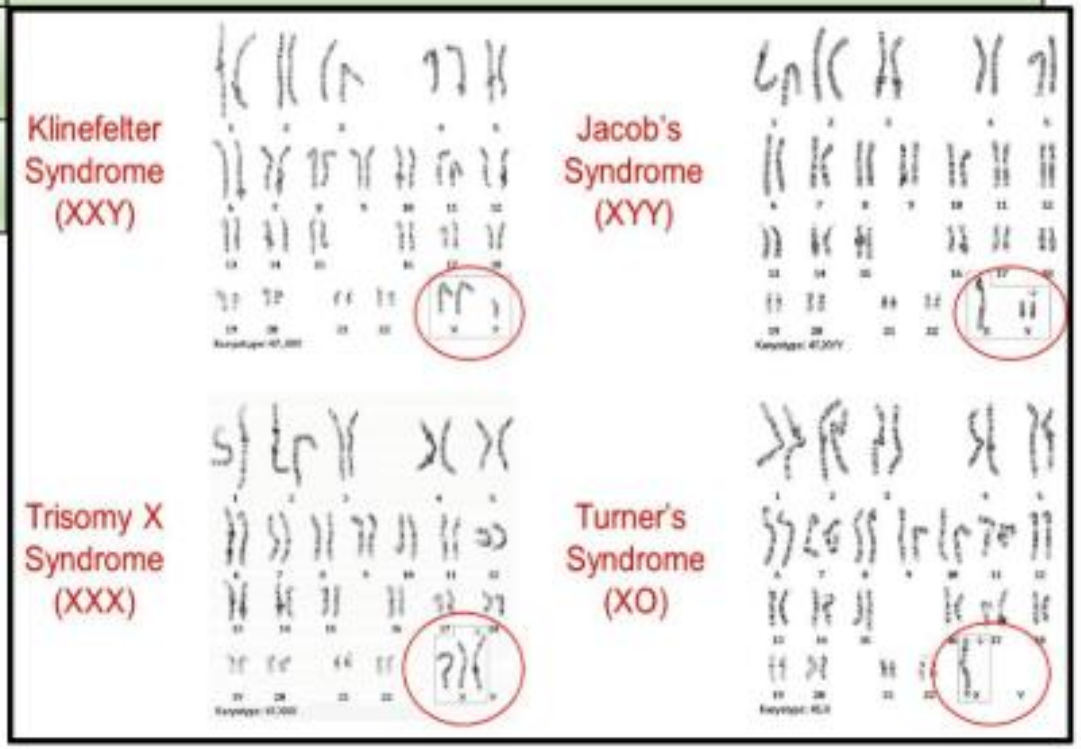
Fertilization ↓

44 + XO **Turner's syndrome**

Disorders due to Non Disjunction of Sex chromosomes

Non-disjunction in Sex chromosomes	Individual's chromosome notation	Physical appearance and symptoms
1) Turner's syndrome	44 + XO	Women with one X-chromosome, short height, webbed neck, lack of underarm and pubic hair, underdeveloped ovaries
2) Klinefelter's syndrome	44 + XXY	Symptoms in males include like tall height, low IQ scores, speech and language difficulty, sterility
3) Jacob's Syndrome	44 + XYY	
4) Trisomy X syndrome	44 + XXX	





Non-disjunction in "Autosomes"	notation	Symptoms
1) Trisomy 18 (Edward's Syndrome)	45 + XX or 45 + XY (extra copy of chromosome 18	Slow growth before birth, low birth weight, abnormal organ growth, heart defects, small head, 90-95% mortality rate during the first year.
2) Trisomy 13 (Patau syndrome)	45 + XX or 45 + XY (extra copy of chromosome 13	Heart defects, abnormalities of the eyes, ears, brain and spinal cord, cleft palate and or lip, small head, low IQ scores, speech and language difficulty, sterility
3) Trisomy 21 (Down Syndrome)	45 + XX or 45 + XY (extra copy of chromosome 21	







Sample questions

1) If cell has 46 chromosomes at the beginning of mitosis, then at anaphase there would be a total of

a) 46 chromosomes

b) 92 chromosomes

c) 23 chromatids

d) 46 chromatids

2) If a cell has 46 chromosomes at the beginning of meiosis, then at anaphase I there would be a total of

a) 23 chromatids

b) 46 chromatids

c) 46 chromosomes

d) 92 chromosomes

3) During cell cycle sister chromatids are pulled apart during

a) Metaphase

b) Anaphase

c) Prophase

d) Interphase

4) The minimum number of meiotic division to obtain 100 sperms

a) 25
b) 50
c) 75
d) 100

5) How many generations and mitotic divisions are required to produce 128 cells ?

a) 6 and 126
b) 7 and 127
c) 5 and 125
d) 4 and 124

6). **Homologous chromosome include**

a) one smaller and one bigger chromosome
b) one chromosome from each parent
c) one complete and one incomplete chromosome
d) none of the above

7) T/F: Interphase is usually divided into 3 phases: G1, S, G2.

8) The _____ is the regular sequence of growth and division that cells undergo.

9) _____ is the stage of the cell cycle where the cell grows to its mature size, copies it DNA, and prepare to divide.

10) Cells can not get too big because:

A. there is not enough DNA to support large cells

B. diffusion is too slow to provide for large cells

C. the surface area of a cell increases too fast for the cell membrane to meet its needs.

D. all of the above

11) DNA is replicated during:

A. interphase

B. prophase

C. metaphase

D. cytokinesis